Chapter 1

INTRODUCTION

Chapter 1: introduction

Our goal:

- get "feel" and terminology
- more depth, detail *later* in course
- approach:
  - use Internet as example

Overview:

- What is the Internet?
- What is a protocol?
- network edge; hosts, access net, physical media
- network core: packet/circuit switching, Internet structure
- performance: loss, delay, throughput
- security
- protocol layers, service models
- history

Chapter 1: roadmap



1.1 what *is* the Internet?

1.2 network edge

- end systems, access networks, links

1.3 network core

- packet switching, circuit switching, network structure

1.4 delay, loss, throughput in networks

1.5 protocol layers, service models

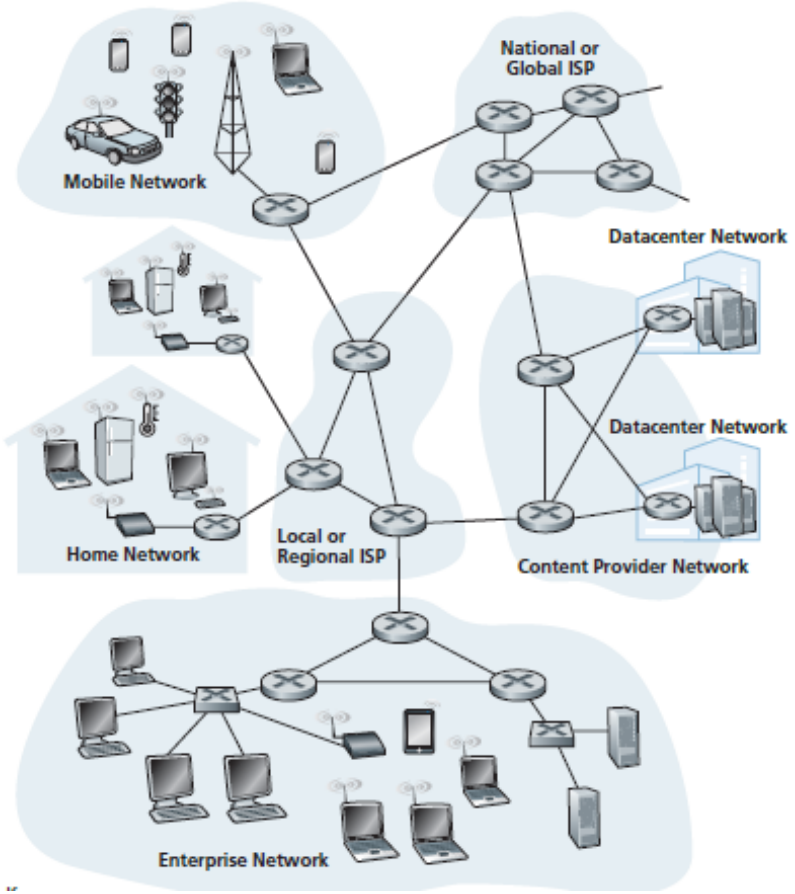
1.6 networks under attack: security

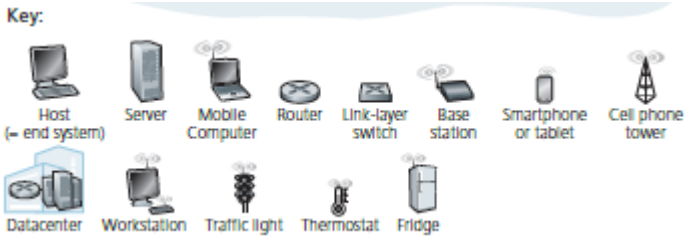
1.7 history

What Is the Internet?

- Internet can be defined in two ways
  - The basic components – i.e. the nuts and bolt
    - the basic hardware and software components that make up the Internet
  - The services it provide
    - a networking infrastructure that provides services to distributed applications

Internet: the network of networks





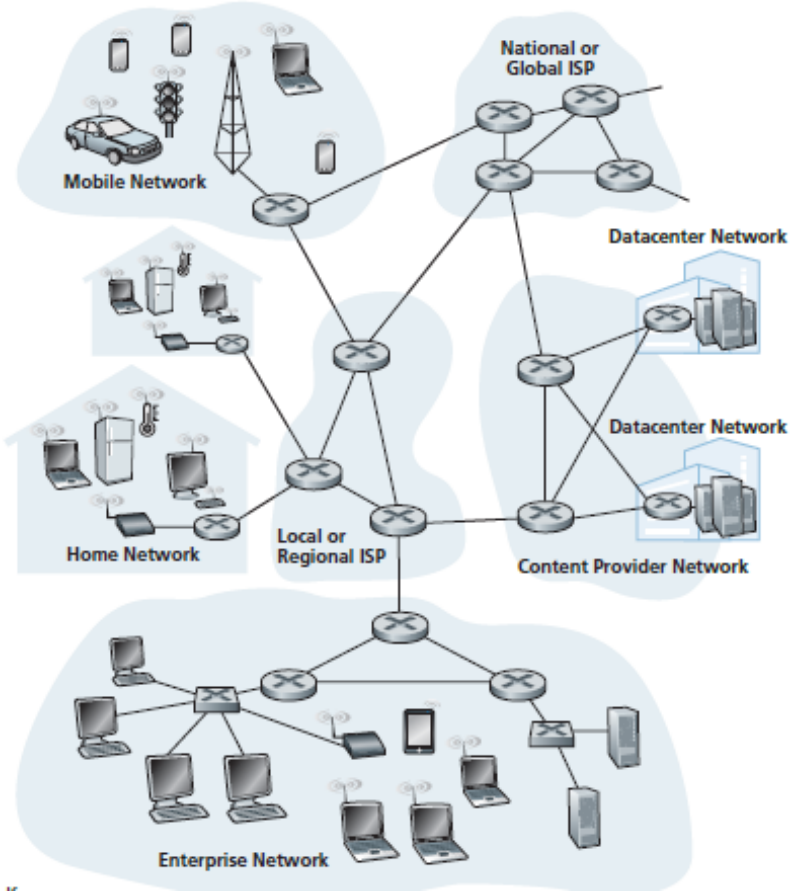
- Billions of connected computing devices:
  - *hosts* = *end systems*
  - running *network apps*

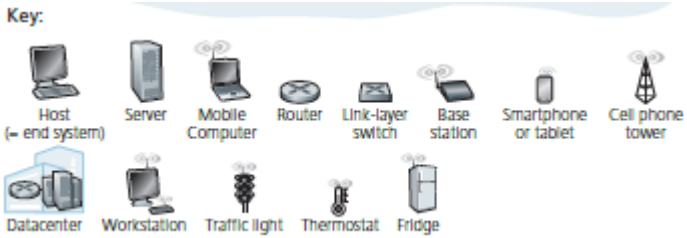
  Internet's "edge"

- *communication links*
  - fiber, copper, radio, satellite
  - transmission rate: *bandwidth*

- *Packet switches:* forward packets (chunks of data)
  - *routers* and *switches*

- *Networks*
  - collection of devices, routers, links: managed by an organization





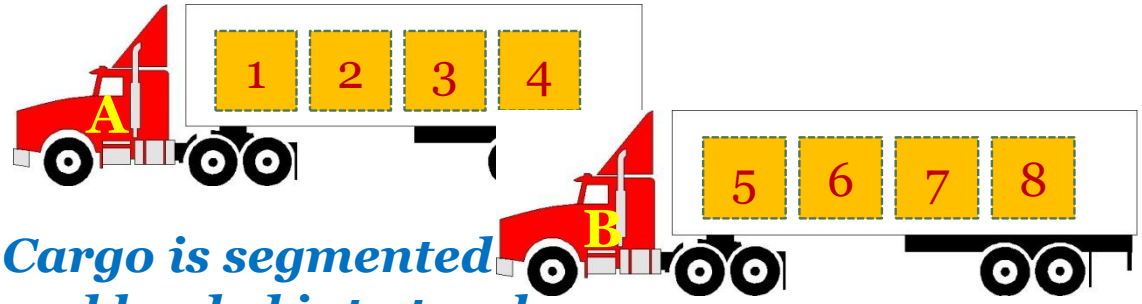
- *Internet:* "network of networks"
  - Interconnected ISPs

protocols are *everywhere*
control sending, receiving of messages
e.g., HTTP (Web), streaming video, Skype, TCP, IP, WiFi, 4G, Ethernet

•All the pieces of the Internet run *protocols* that controls the sending and receiving of information
•TCP/IP are two of the most important protocols in the Internet.

- *Internet standards*
  - RFC: Request for Comments
  - IETF: Internet Engineering Task Force

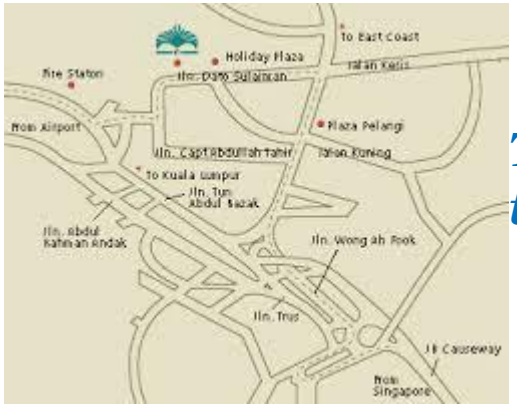
Packet-switched networks (analogy)



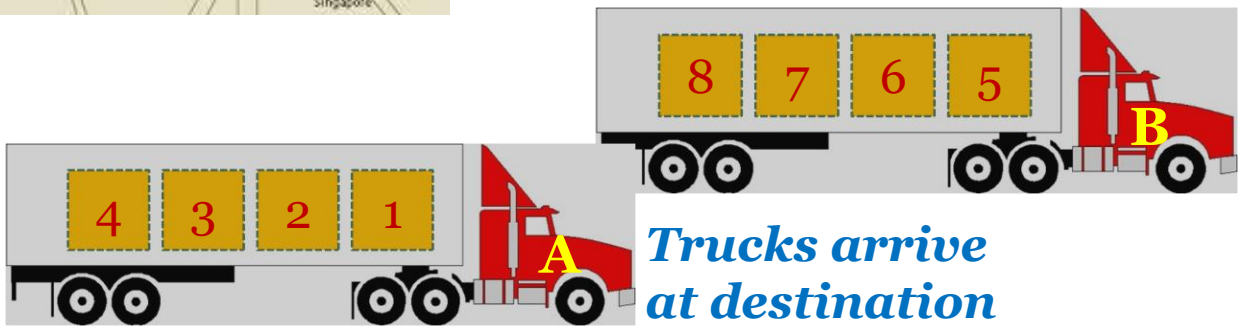
Cargo is segmented and loaded into trucks

Warehouse

1	2	3	4
5	6	7	8



Trucks go on their way

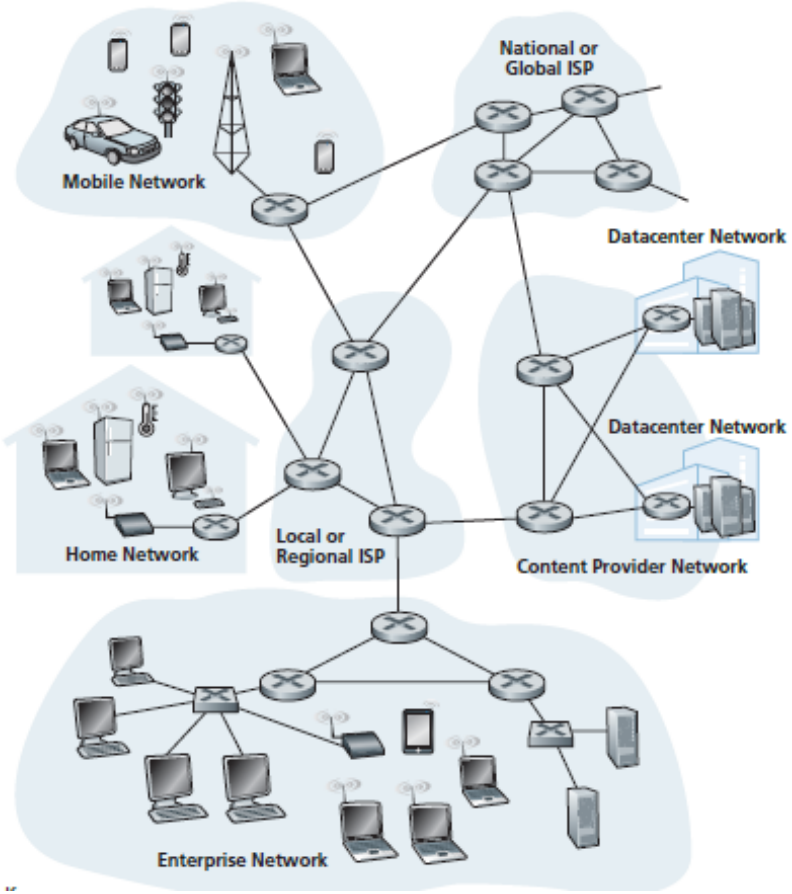


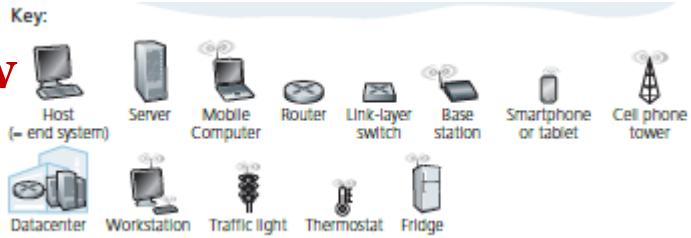
Trucks arrive at destination

Cargo is unloaded and grouped

1	2	3	4
5	6	7	8

The Internet: a "services" view

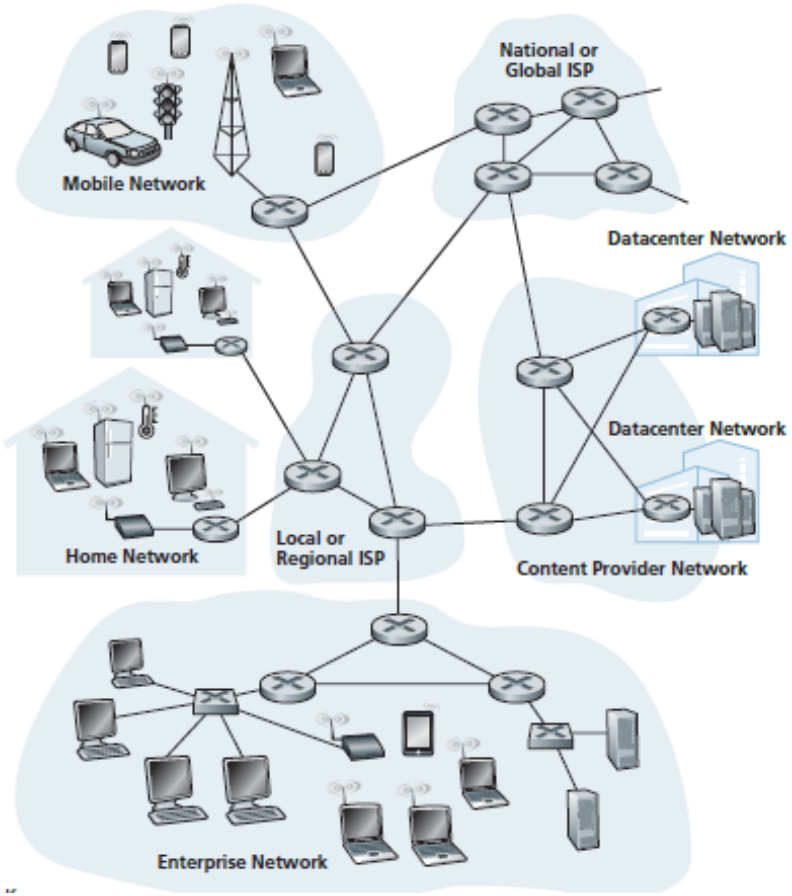


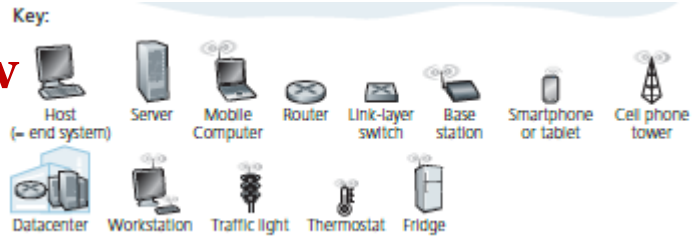


- *Infrastructure* that provides services to applications:
  - Web, streaming video, multimedia teleconferencing, email, games, e-commerce, social media, inter-connected appliances, …

- provides *programming interface* to distributed applications:
  - "hooks" allowing sending/receiving apps to "connect" to, use Internet transport service
  - provides service options, analogous to postal service

The Internet: a "services" view





- *Distributed applications* ➔ involve multiple end systems exchanging data
  - Have *Application Programming Interface (API)*
  - to communicate between different platforms through the Internet
- Internet is a network of networks
  - Interconnected ISPs

“Fun” internet appliances



Amazon Echo

Web-enabled toaster + weather forecaster





Pacemaker & Monitor



IP picture frame
http://www.ceiva.com/



Fitbit



Tweet-a-watt: monitor energy use



AR devices



Internet phones



Internet refrigerator



sensorized, bed mattress



Gaming devices

bikes





cars



Slingbox: watch, control cable TV remotely



Security Camera

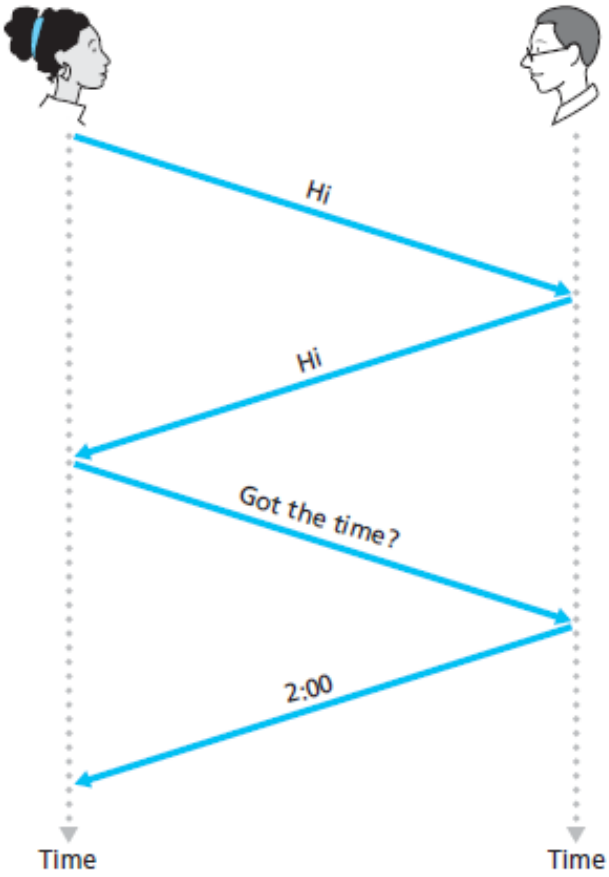


scooters

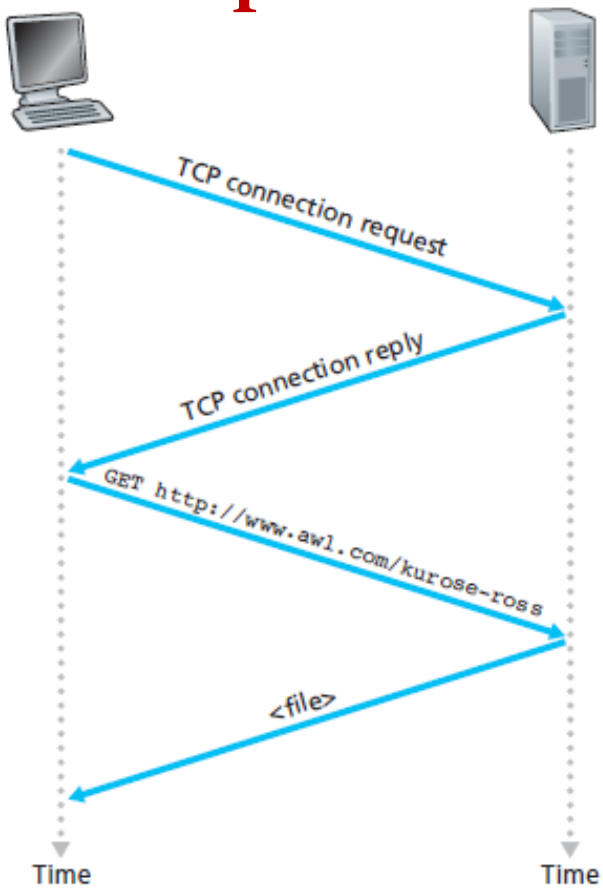
What Is a Protocol?

- **Protocol** is a set of rules governing the exchange or transmission of data between devices.
  - defines format, order of messages exchanged (sent and received) among network entities, and actions taken on message transmission, receipt or other event
  - e.g. TCP, IP, HTTP, Skype, 802.11
- Internet standards
  - RFC: Request for comments
  - IETF: Internet Engineering Task Force

A human protocol



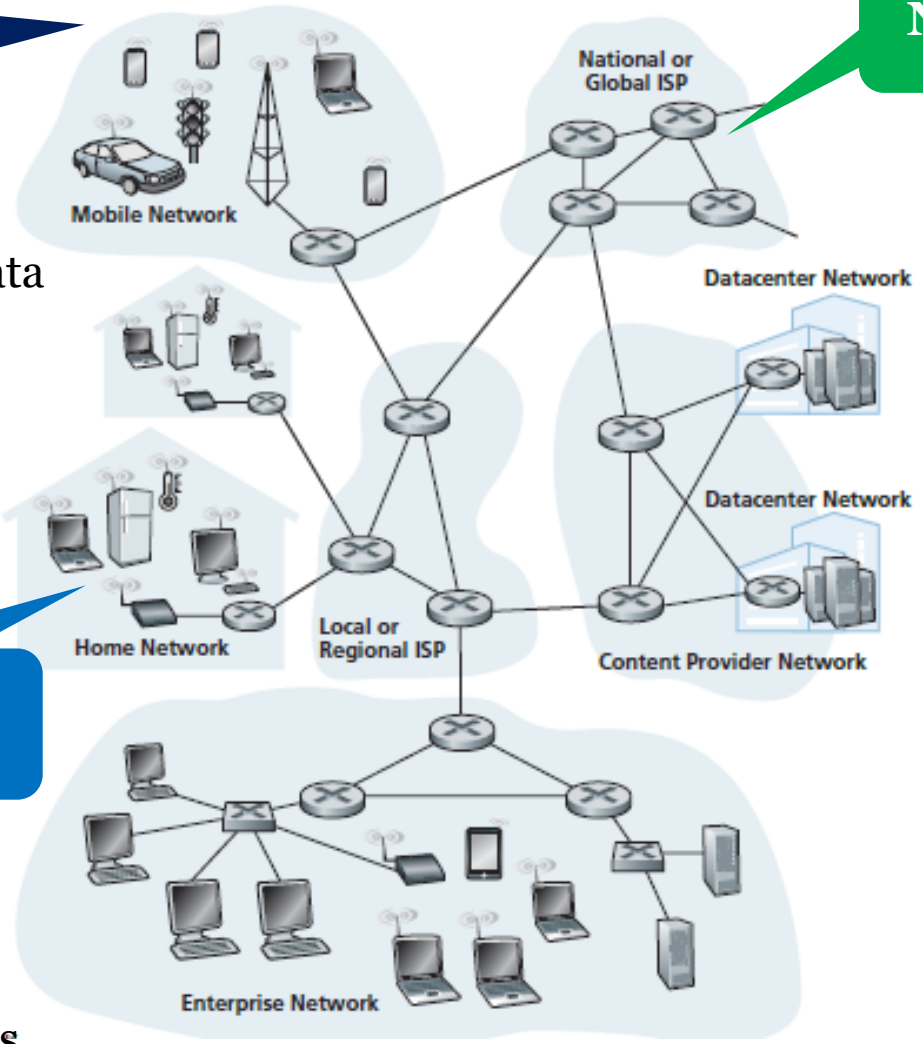
A computer network protocol



Network edge

Network edge:

- hosts: clients and servers
- servers often in data centers



Network Core

Network core:

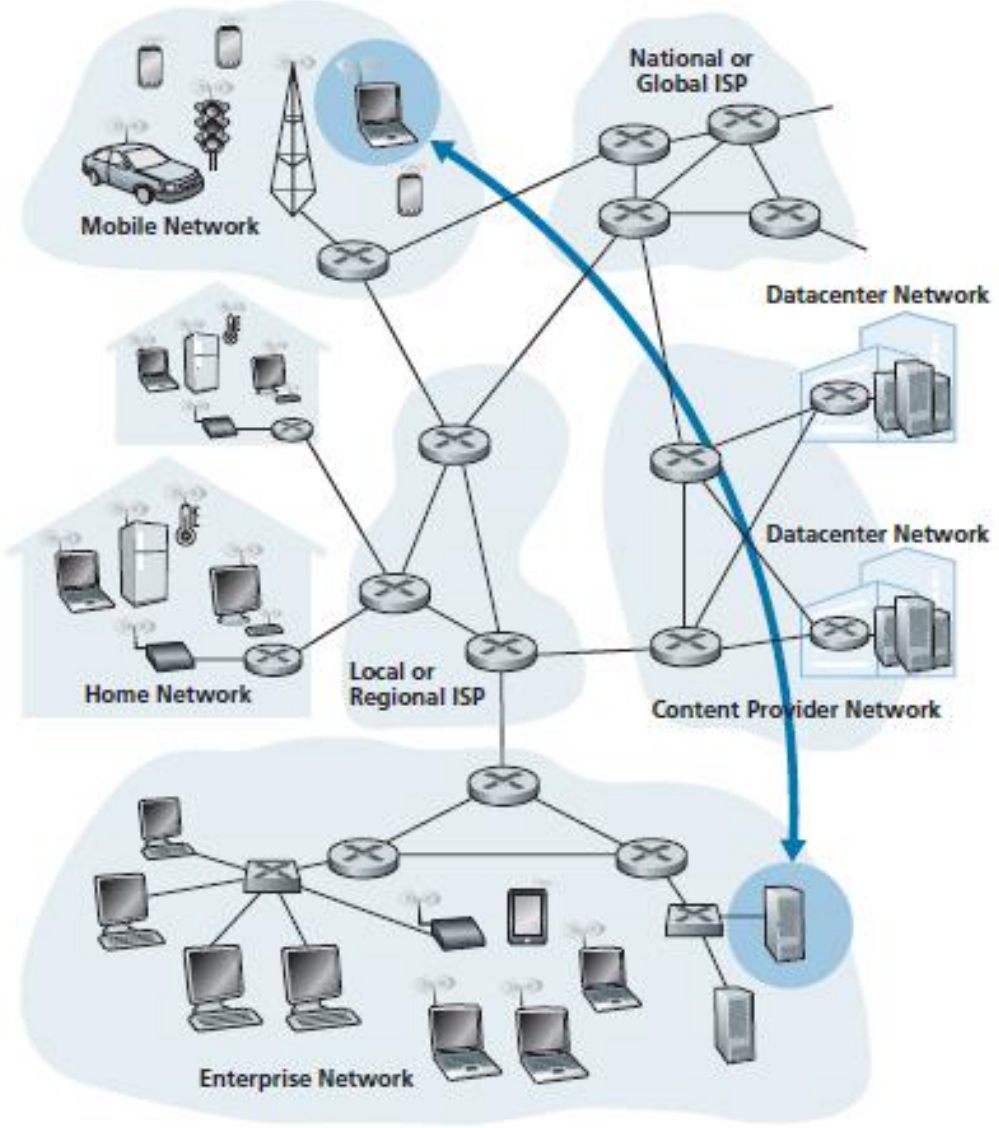
interconnected routers

network of networks

Access Networks

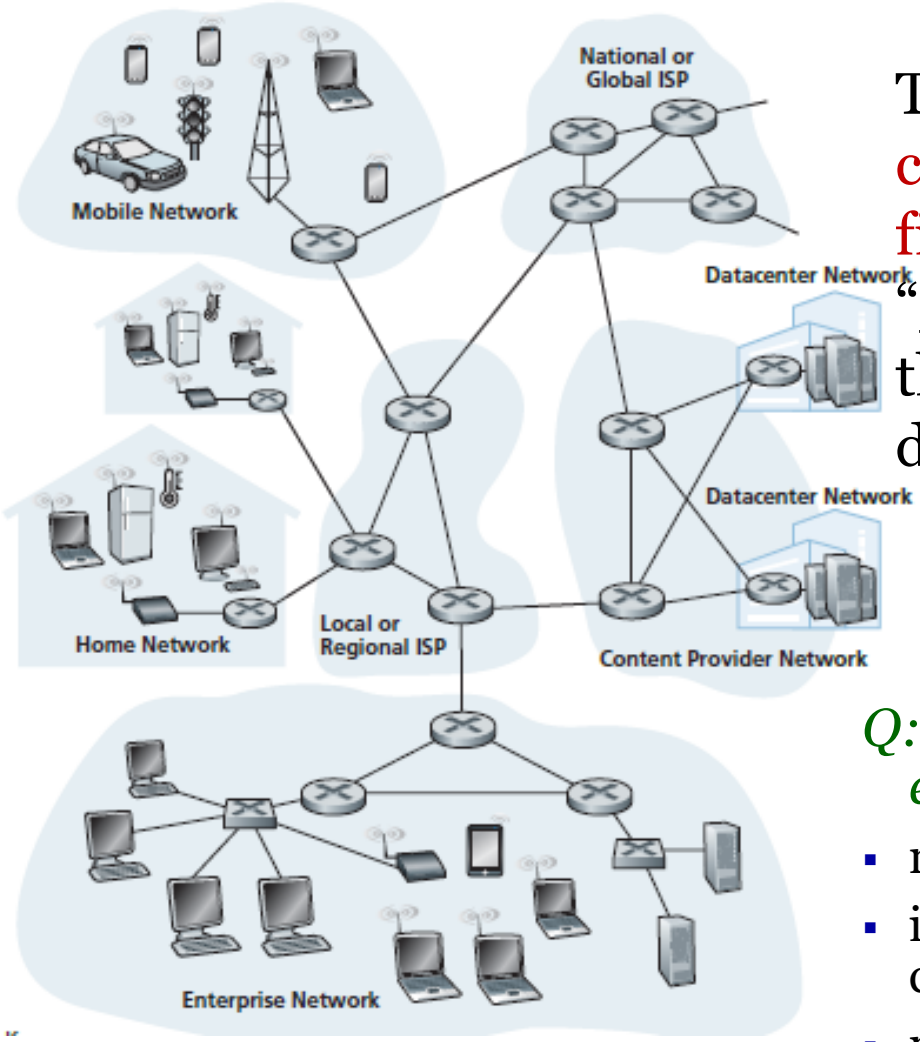
Access networks, physical media:

wired, wireless communication links



End system interaction

Access networks and physical media



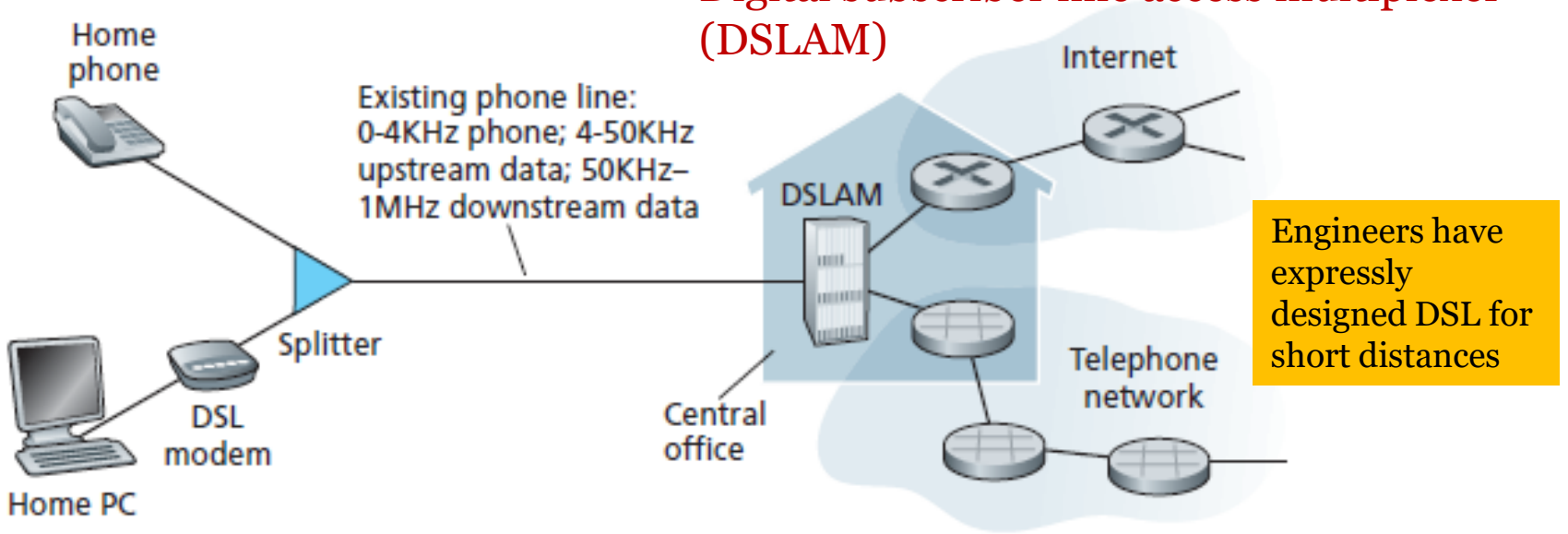
The network that physically connects an end system to the first router (also known as the "*edge router*") on a path from the end system to any other distant end system.

Q: How to connect end systems to edge router?

- residential access networks
- institutional access networks (school, company)
- mobile access networks (WiFi, 4G/5G)

Access Network : DSL

Digital subscriber line access multiplexer (DSLAM)

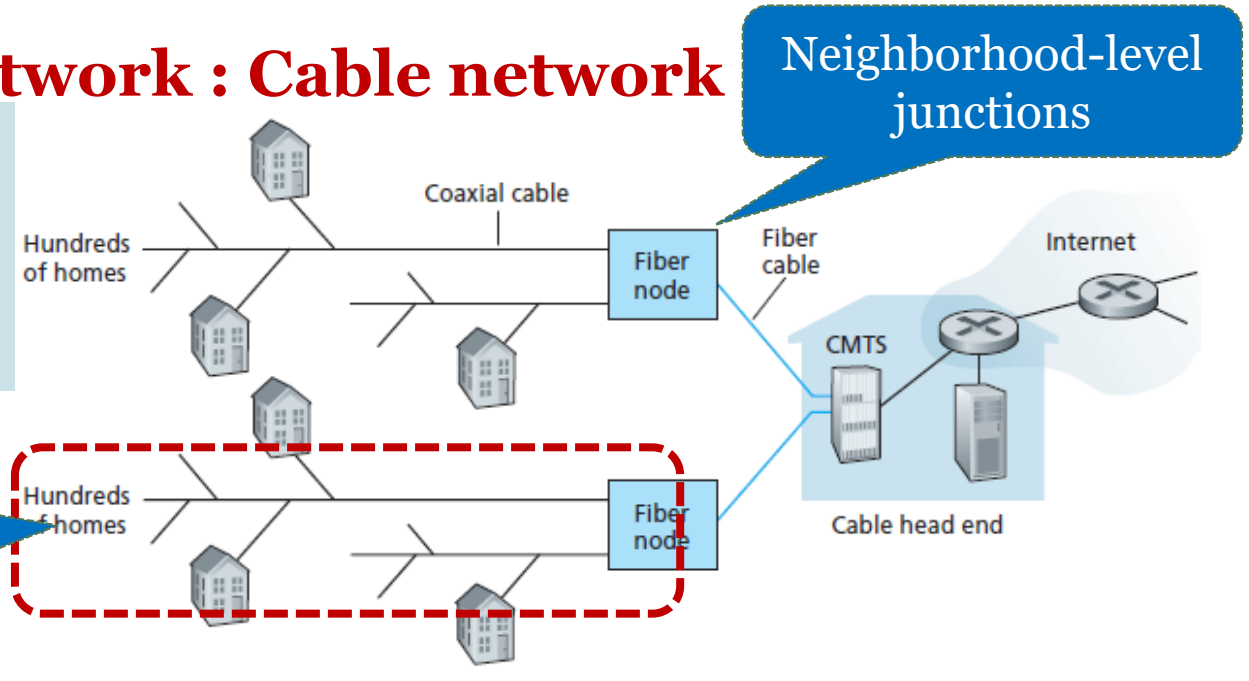


Engineers have expressly designed DSL for short distances

- A customer's telco is also its ISP.
- The DSL modem uses the existing telephone line to exchange data with a DSLAM located in the telco's local central office (CO).
  - < 2.5 Mbps upstream transmission rate (typically < 1 Mbps)
  - < 24 Mbps downstream transmission rate (typically < 10 Mbps

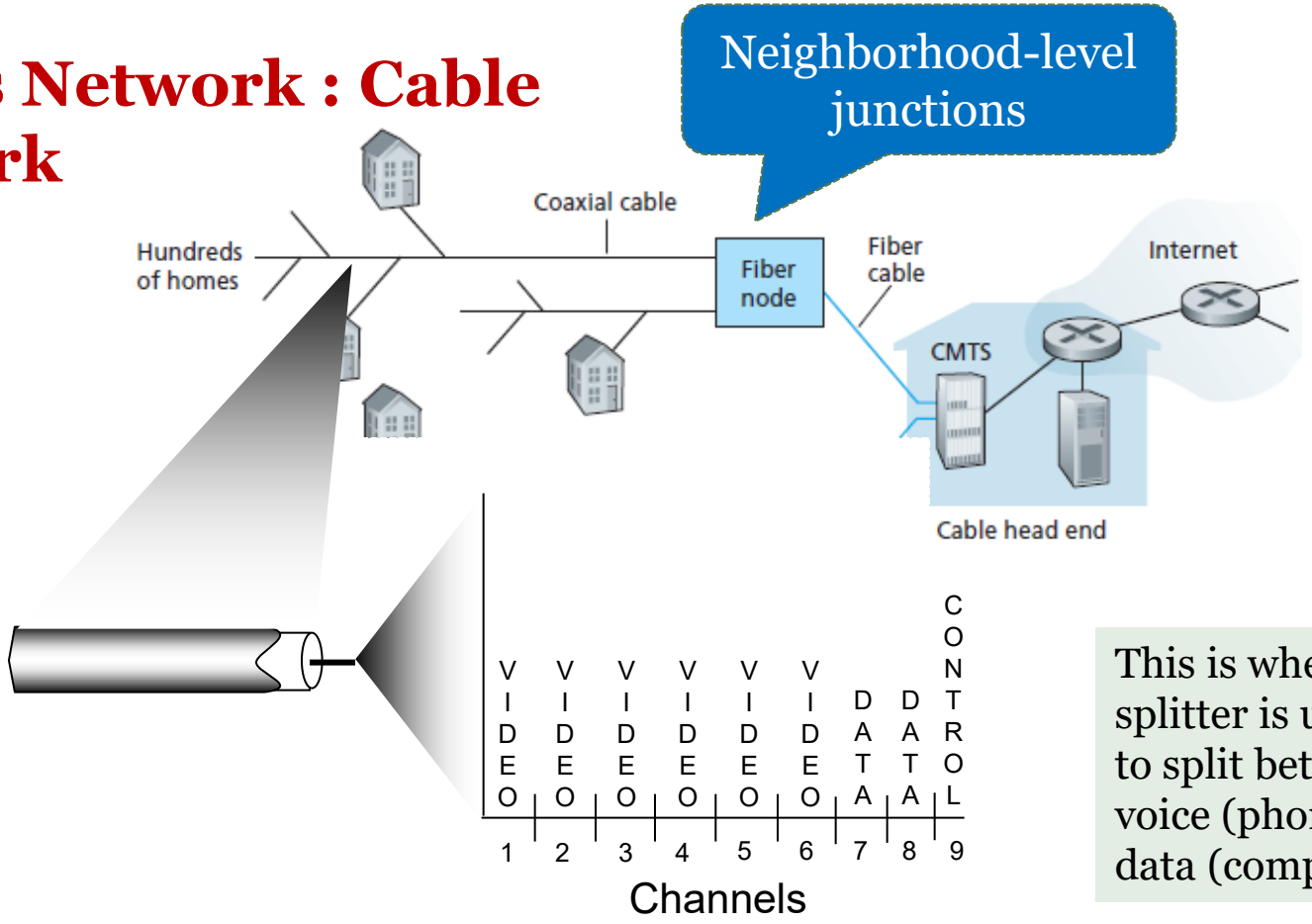
Access Network : Cable network

data, TV transmitted at different frequencies over shared cable distribution network



- Cable Internet access makes use of the cable television company's existing cable television infrastructure.
  - A hybrid fiber-coaxial (HFC) access network
  - asymmetric: up to 30Mbps downstream transmission rate, 2 Mbps upstream transmission rate
- network of cable, fiber attaches homes to ISP router
  - homes share access network to cable headend
  - unlike DSL, which has dedicated access to central office

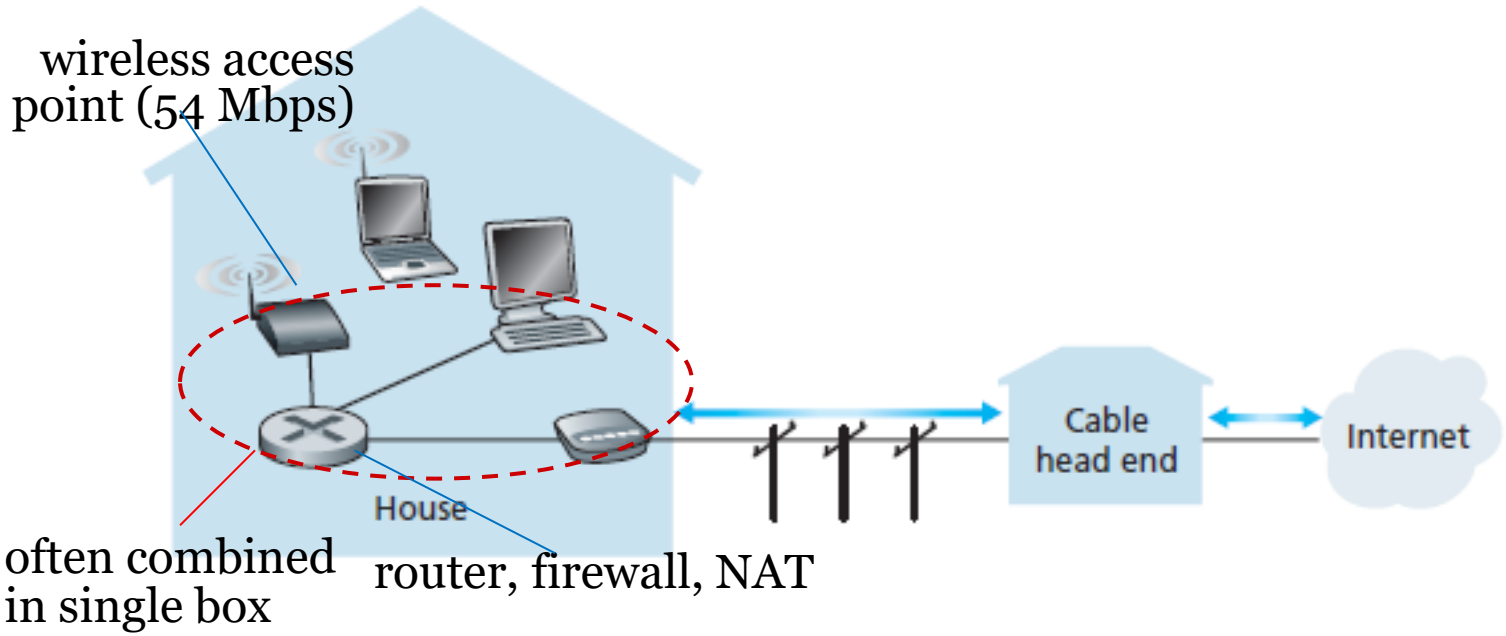
Access Network : Cable network



This is where the splitter is useful – to split between voice (phone) and data (computer)

frequency division multiplexing: different channels transmitted in different frequency bands

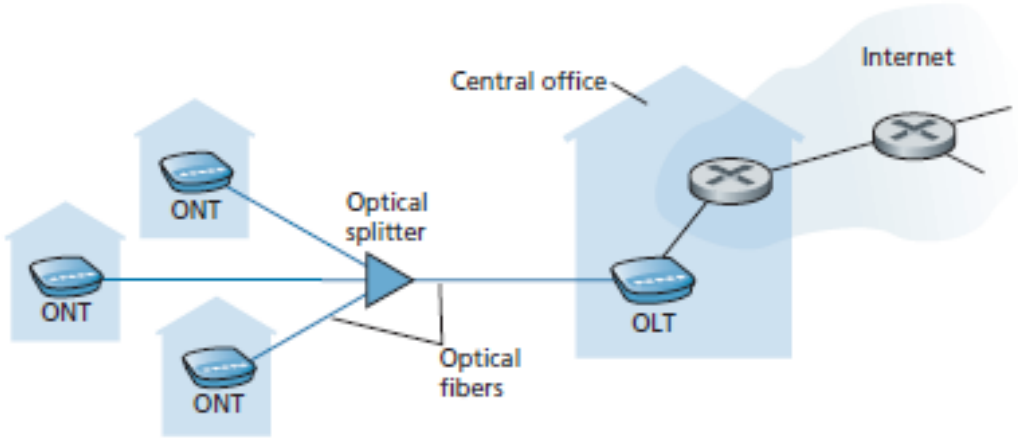
Access net: home network



•Many homes combine broadband residential access (that is, cable modems or DSL) with these inexpensive wireless LAN technologies to create powerful home networks

FTTH Internet access

up-and-coming technology



- An up-and-coming technology that provides even higher speeds is **fiber to the home (FTTH)** [Fiber Broadband 2020].
- The FTTH concept is simple—provide an optical fiber path from the Central Office directly to the home.
- Can potentially provide Internet access rates in the ***gigabits per second range***.

How FTTH Works

- **Optical Fiber Transmission**: FTTH uses **glass or plastic fibers** to transmit data as **light pulses**, which is much faster and more efficient than traditional copper cables.
- **Direct Connection**: Unlike older systems (DSL or cable), which share bandwidth among users in a neighborhood, FTTH gives each home its **own dedicated fiber line**.
- **Optical Network Terminal (ONT)**: Inside the home, an ONT converts the light signals into digital signals that your devices can use.

Key Benefits of FTTH

Feature	FTTH Advantage
Speed	Gigabit speeds (1 Gbps or more) for downloads and uploads
Reliability	Less interference and signal degradation compared to copper
Latency	Extremely low latency—ideal for gaming, video conferencing, and real-time applications
Future-Proofing	Can support increasing bandwidth demands for decades
Streaming & Downloads	Seamless 4K/8K streaming, instant cloud access, and rapid file transfers

5G fixed wireless

up-and-coming technology

5G Fixed Wireless Access (FWA) is a ***broadband solution*** that delivers *high-speed internet to homes and businesses* using 5G cellular technology, without the need for physical cables like fiber or copper

- **5G fixed wireless** is beginning to be deployed*.
- 5G fixed wireless ➔ promises high-speed residential access, but will do so without installing costly and failure-prone cabling from the telco's central office (CO) to the home.
- With 5G fixed wireless (using ***beam-forming technology***) data is sent wirelessly from a provider's base station to a modem in the home.

How It Works

Component	Role
5G Base Station	Sends data wirelessly using 5G signals
Beamforming Technology	Directs focused radio waves to specific homes for stronger, more reliable connections
Home Modem/Receiver	Captures the 5G signal and converts it into usable internet for devices

Instead of laying fibre from the central office (CO) to each home, the provider installs a **receiver or modem** at the customer's location that connects wirelessly to the nearest 5G tower.

Key Benefits

- **No Cabling Needed**: Eliminates the cost and complexity of underground fibre installation
- **High Speeds**: Comparable to fibre in many cases—often hundreds of Mbps to over 1 Gbps
- **Quick Deployment**: Faster rollout, especially in rural or underserved areas
- **Mobility-Friendly**: Can support both fixed and mobile broadband needs

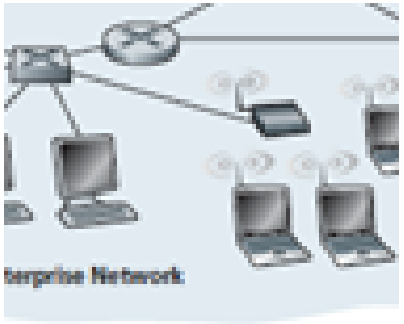
Feature	FTTH	5G Fixed Wireless
Connection Type	Wired (fiber optic)	Wireless (radio waves)
Installation	Requires physical cabling	No cabling—uses towers
Speed	Very high (1–10 Gbps)	High (100 Mbps–1 Gbps)
Latency	Very low	Low, but can vary
Scalability	Excellent	Good, but depends on spectrum and tower density
Best For	Dense urban areas, long-term infrastructure	Rapid deployment, rural or suburban areas

Wireless access networks

• Shared wireless access network connects end system to router via base station A.K.A. "access point"

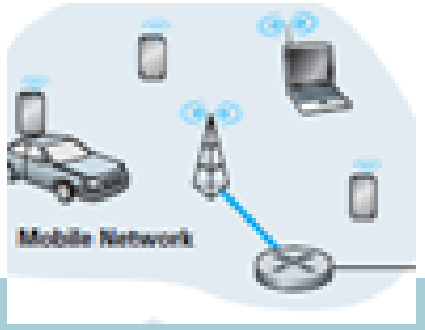
wireless LANs:

- within building (~100 ft: 30.5 metres)
- 802.11b/g/n (WiFi): 11, 54, 450 Mbps transmission rate

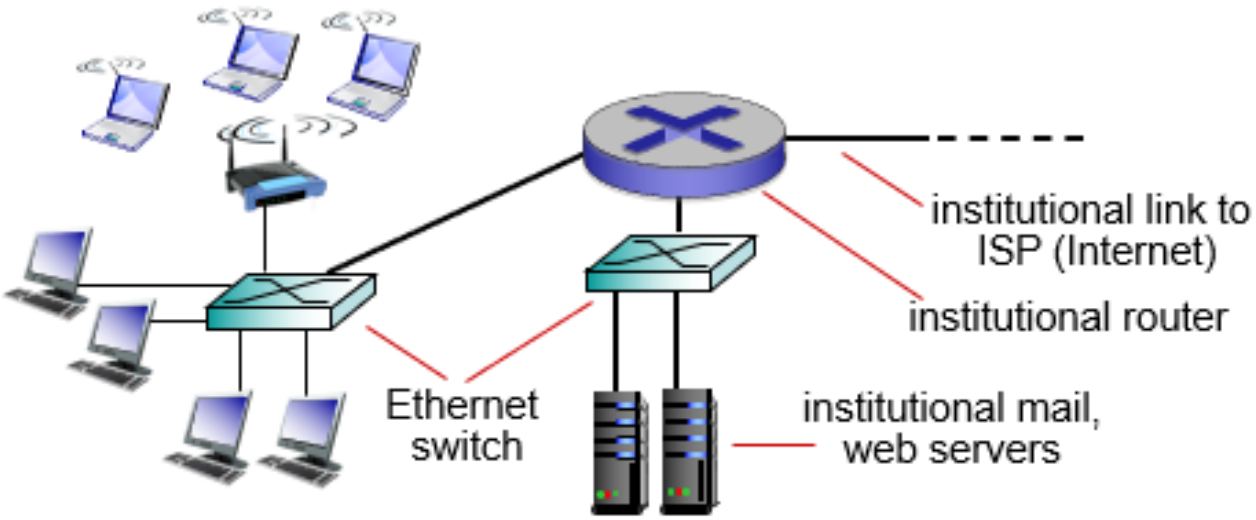


wide-area cellular access networks

- provided by mobile (cellular) operator, 10's km
- between 1 and 10 Mbps
- 4G/5G cellular networks



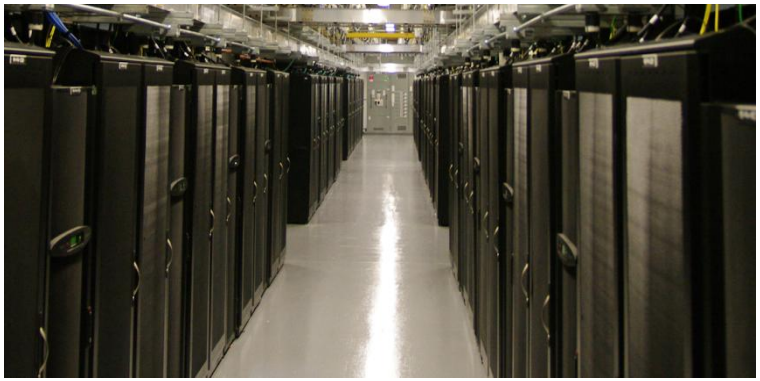
Enterprise Access networks (Ethernet)



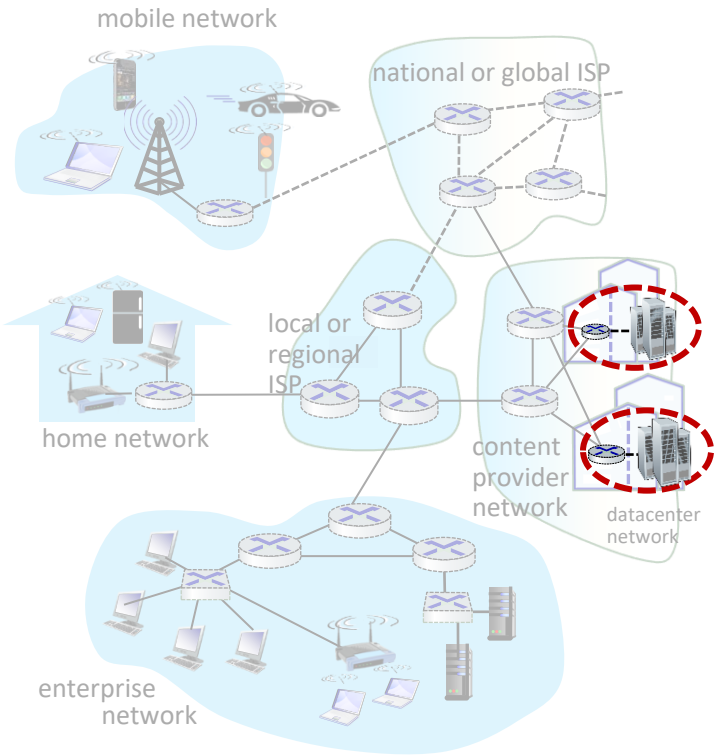
- Typically used in companies, universities, schools, etc.
- 10 Mbps, 100Mbps, 1Gbps, 10Gbps transmission rates
- A mixed of wired and wireless link technologies/
  - Ethernet: wired access at 100Mbps, 1Gbps, 10Gbps
  - WiFi: wireless access points at 11, 54, 450 Mbps

Access networks: data center networks

- high-bandwidth links (10s to 100s Gbps) connect hundreds to thousands of servers together, and to Internet



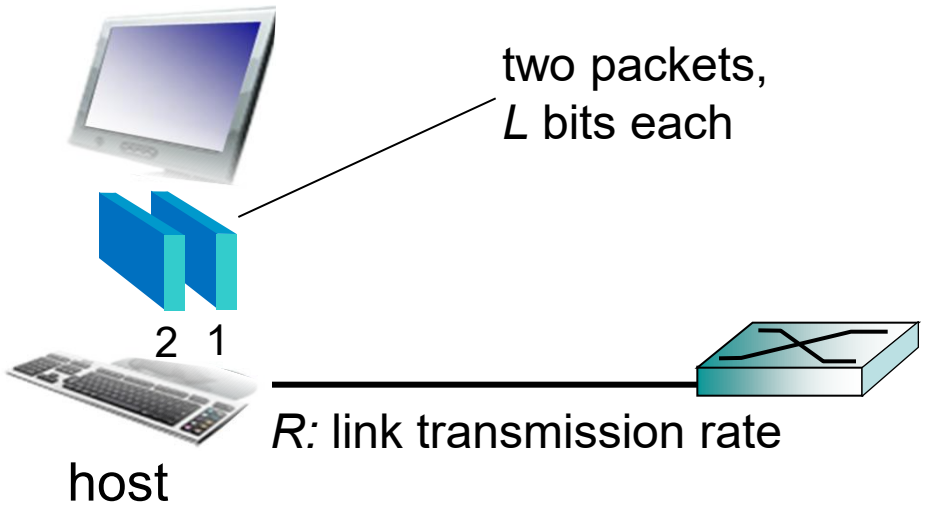
Courtesy: Massachusetts Green High Performance Computing Center (mghpcc.org)



Host: sends *packets* of data

host sending function:

- takes application message
- breaks into smaller chunks, known as *packets*, of length *L* bits
- transmits packet into access network at *transmission rate R*
  - link transmission rate, aka link *capacity, aka link bandwidth*



$$\text{packet transmission delay} = \text{time needed to transmit } L\text{-bit packet into link} = \frac{L \text{ (bits)}}{R \text{ (bits/sec)}}$$

packet transmission delay = time needed to transmit *L*-bit packet into link

$$= \frac{L \text{ (bits)}}{R \text{ (bits/sec)}}$$

Example: L=30Mbit, R=10Mbps

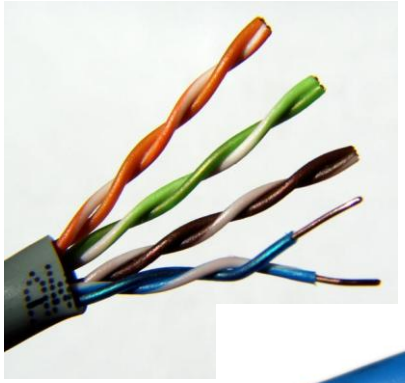
packet transmission delay
= L/R
= 30Mbit/10Mbps
= 3s

Physical Media: The links

- Bit: propagates between transmitter/receiver pairs
- physical link: what lies between transmitter & receiver
- Physical media fall into two categories:
  - **Guided media**
    - signals propagate in solid media
    - E.g.: twisted-pair copper wire, coaxial cable, multimode fiber-optic cable
  - **Unguided media.**
    - signals propagate freely, e.g., radio
    - E.g.: terrestrial radio spectrum, and satellite radio spectrum.

Twisted pair (TP)

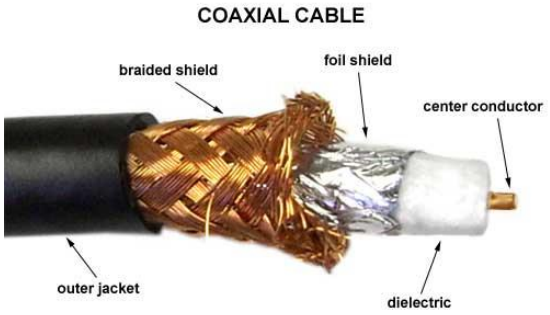
- two insulated copper wires
  - Category 5: 100 Mbps, 1 Gbps Ethernet
    - Category 6: 10Gbps



Coaxial cable:

- two concentric copper conductors
- bidirectional
- broadband:
  - multiple frequency channels on cable
  - 100's Mbps per channel
  - HFC







Hybrid Fibre Coaxial (HFC) broadband is a type of internet service that combines fiber optic and coaxial cables to deliver high-speed data

Fiber optic cable:

- glass fiber carrying light pulses, each pulse a bit
- high-speed operation:
  - high-speed point-to-point transmission (e.g., 10's-100's Gpbs transmission rate)
- low error rate:
  - repeaters spaced far apart
  - immune to electromagnetic noise





Wireless Radio

- signal carried in electromagnetic spectrum
- no physical "wire"
- Bidirectional
  - broadcast, "half-duplex" (sender to receiver)
- propagation environment effects:
  - reflection
  - obstruction by objects
  - Interference/noise

radio link types:

- terrestrial microwave
  - Point-to-point; up to 45 Mbps channels
- Wireless LAN (WiFi)
  - 10-100's Mbps, 10's meters
- wide-area (e.g., 4G cellular)
  - 10 Mbps over ~ 10km
- Bluetooth: cable replacement
  - short distances, limited rates
- satellite
  - Up to 45Mbps per channel
  - 270 msec end-end delay
  - Two types of satellites are used in communications: geostationary satellites and low-earth orbiting (LEO) satellites.

Chapter 1: roadmap



1.1 what *is* the Internet?

1.2 network edge

- end systems, access networks, links

1.3 network core

- Packet/circuit switching, internet structure

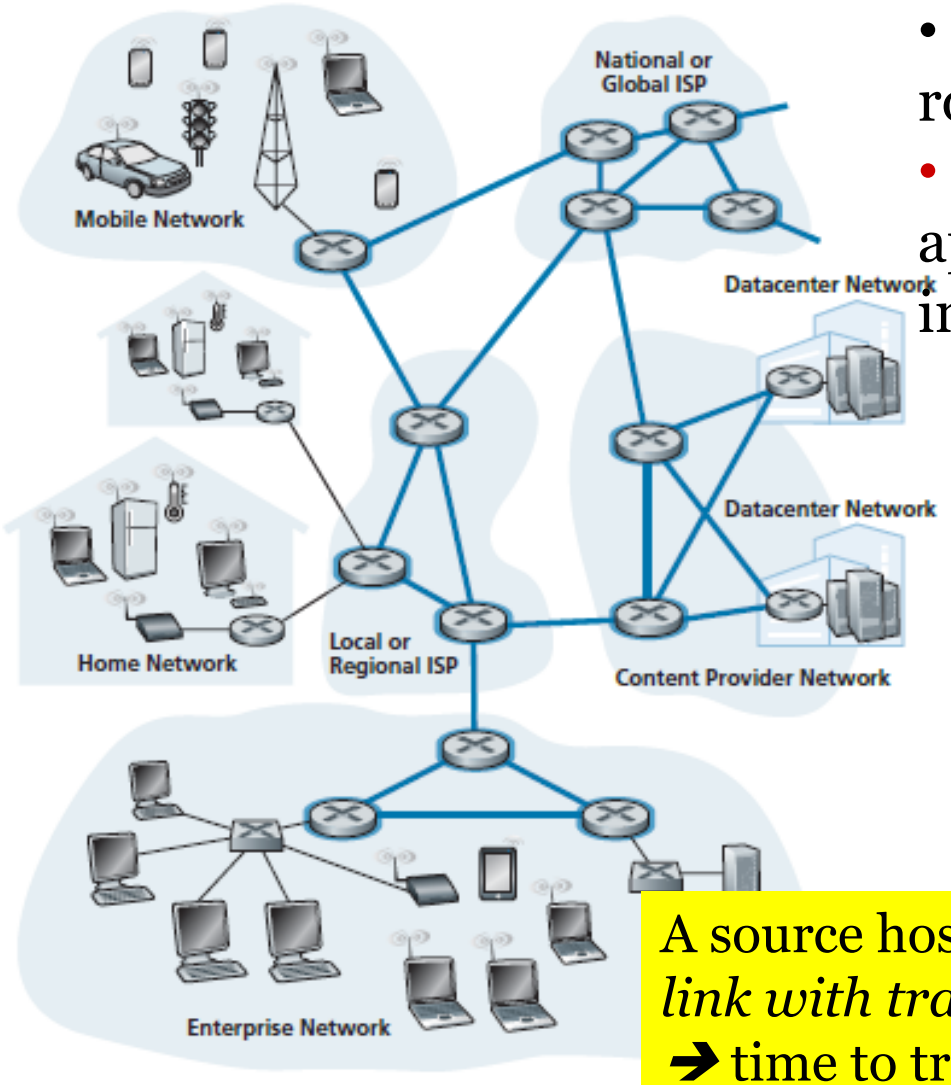
1.4 delay, loss, throughput in networks

1.5 protocol layers, service models

1.6 networks under attack: security

1.7 history

The network core



- Mesh of interconnected routers
- Packet-switching: hosts break application-layer messages into *packets*
  - Network forward packets from one router to the next, across links on path from source to destination
  - Each packet transmitted at full link capacity

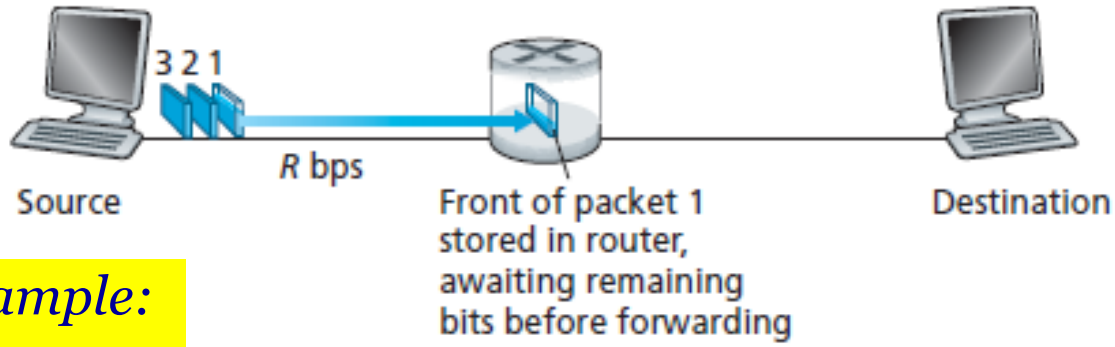
A source host sends a packet of *L bits over a link with transmission rate R bits/sec*
➔ time to transmit the packet is *L/R secs*

Packet Switching: Store-and-Forward Transmission

1 hop = from source to router

- Store-and-forward transmission ➔ the router must receive the entire packet before it can begin to transmit onto the outbound link

Takes L/R seconds to transmit (push out) L-bit packet into link at R bps



one-hop numerical example:

- L = 7.5 Mbits
- R = 1.5 Mbps
- one-hop transmission delay = L/R = 5 sec

end-end delay = $2L/R$ (assuming zero propagation delay)

one-hop numerical example:

- L = 7.5 Mbits
- R = 1.5 Mbps
- one-hop transmission delay = L/R = 5 sec

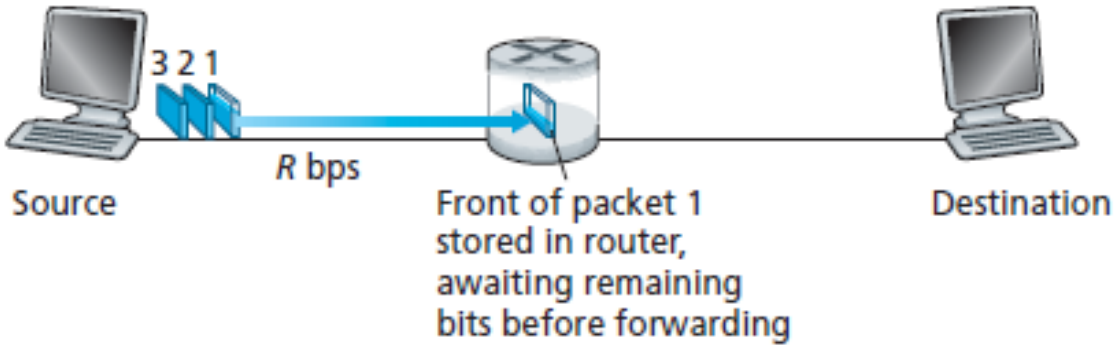
One-hop numerical example:

- L = 10 Kbits
- R = 100 Mbps
- one-hop transmission delay = 0.1 msec

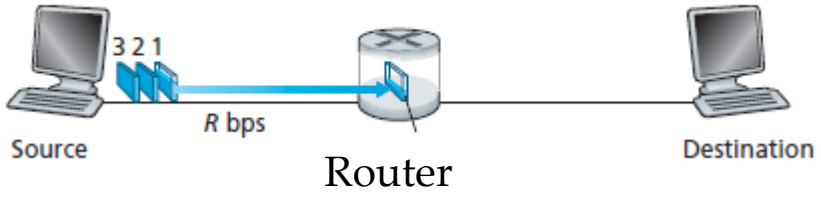
transmitting some of packet 1

the source has three packets, each consisting of *L bits, to send to the destination.*

When the router has received ***all*** *of the packet's bits can it begin to transmit the packet onto the* outbound link.



The router employs store-and-forwarding ➔ so must first buffer ("store") the packet's bits



If not store-n-forward:

end-end delay = *L/R* (assuming no propagation delay)

Time

0	L/R	2L/R	3L/R	4L/R
Source begins transmit Packet	Router received the whole packet	Destination received the whole packet		

end-end delay = 2*L/R* (assuming no propagation delay)

For 3 packets

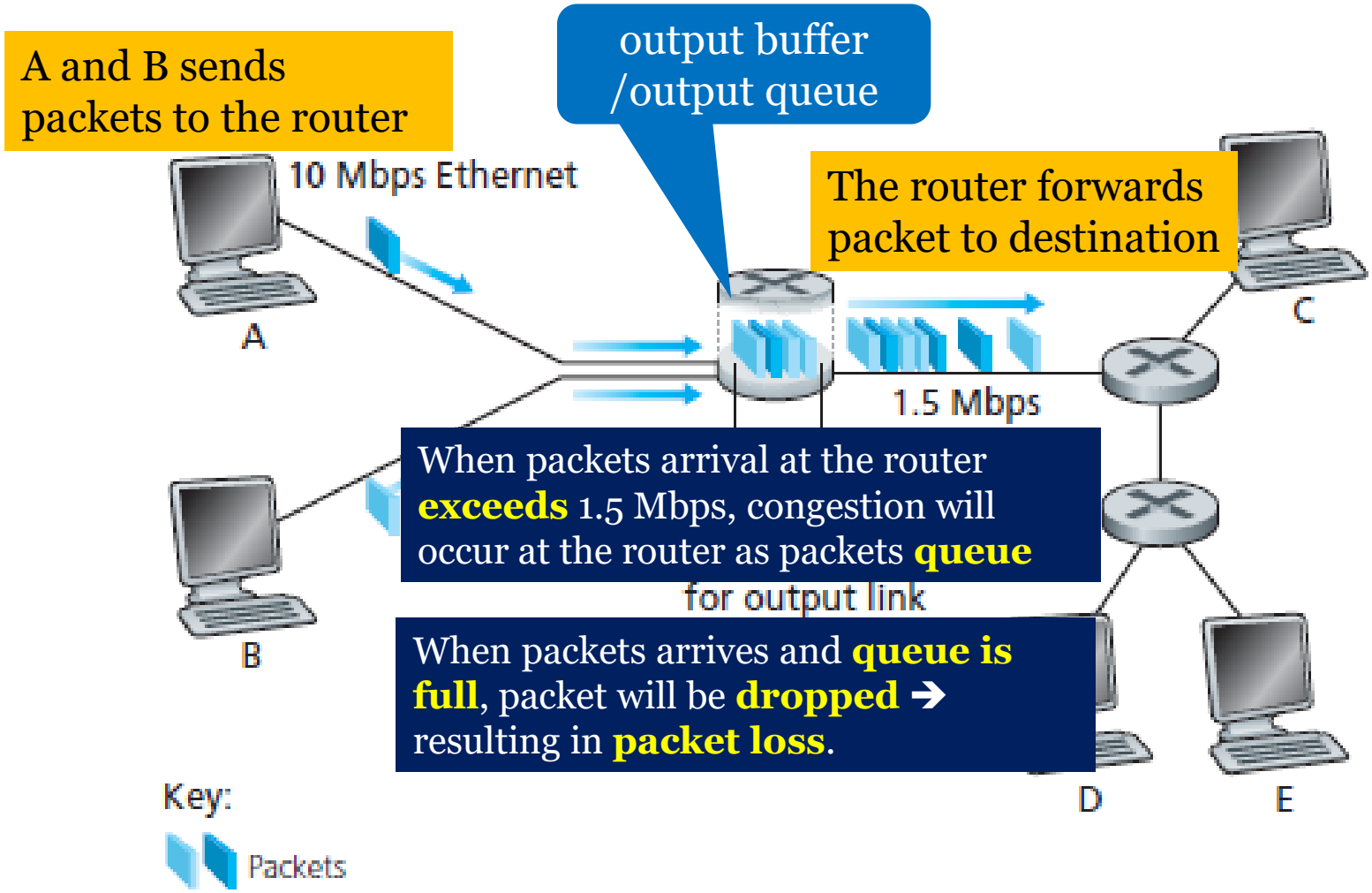
0	L/R	2L/R	3L/R	4L/R
Source begins transmit P#1	Router received the whole packet P#1	Destination received the whole packet P#1	Destination received the whole packet P#2	Destination received the whole packet P#3
	Source begins transmit P#2	Router received the whole packet P#2	Router received the whole packet P#3	
		Source begins transmit P#3		

end-end delay = N(*L/R*)

where

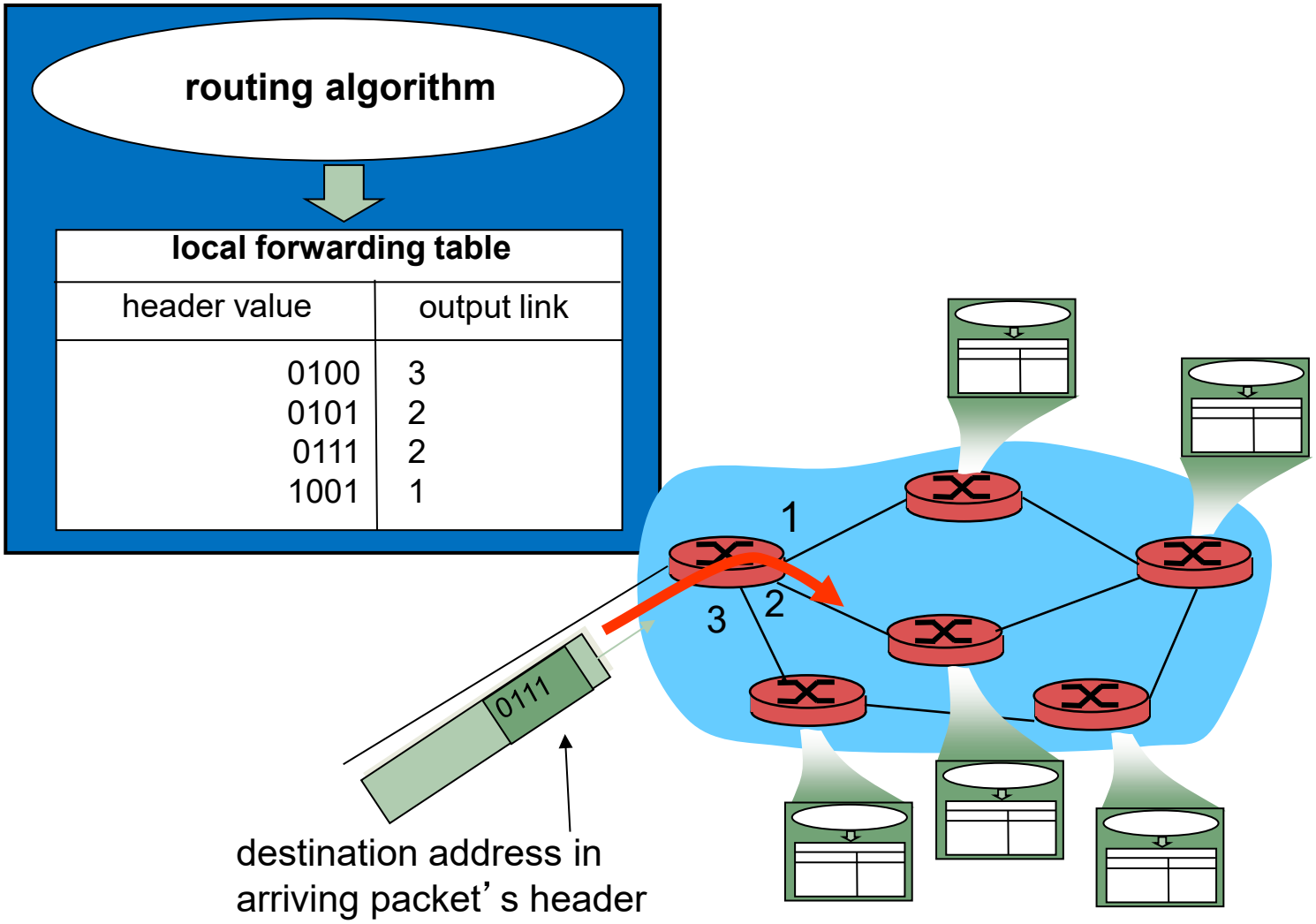
N = the number of links between source and destination

Packet Switching: Queuing Delays and Packet Loss



Two key network-core functions

ROUTING	FORWARDING
• determines source to destination route taken by packets ○ routing algorithms • Global action	• move packets from router input interface to appropriate router output interface • Local Action


routing algorithm
local forwarding table
header value
output link
0100 3
0101 2
0111 2
1001 1
1
2
3
0111
destination address in
arriving packet's header

1 Router R1 receives a packet

2 R1 checks the destination address

3 R1 finds the best path to destination

4 R1 forwards the packet (or drops it)

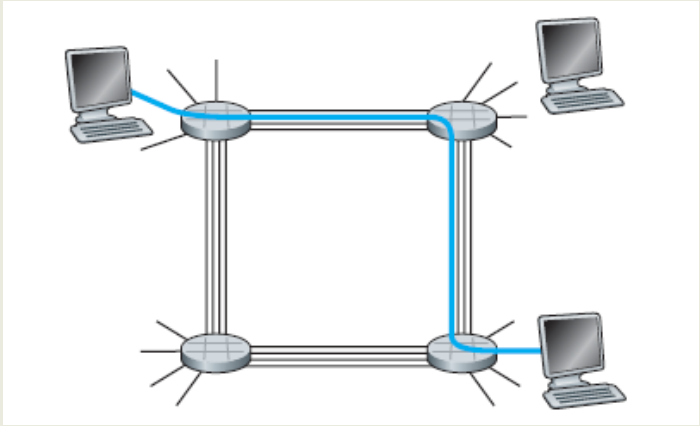
Packet destination: 192.168.1.1

```
R1# show ip route
     172.16.0.0/24 is subnetted, 3 subnets
C       172.16.1.0 is directly connected, FastEthernet0/0
C       172.16.2.0 is directly connected, Serial0/0/0
R       172.16.3.0 [120/1] via 172.16.2.2, 00:00:25, Serial0/0/0
R    192.168.1.0/24 [120/1] via 172.16.2.2, 00:00:25, Serial0/0/0
```

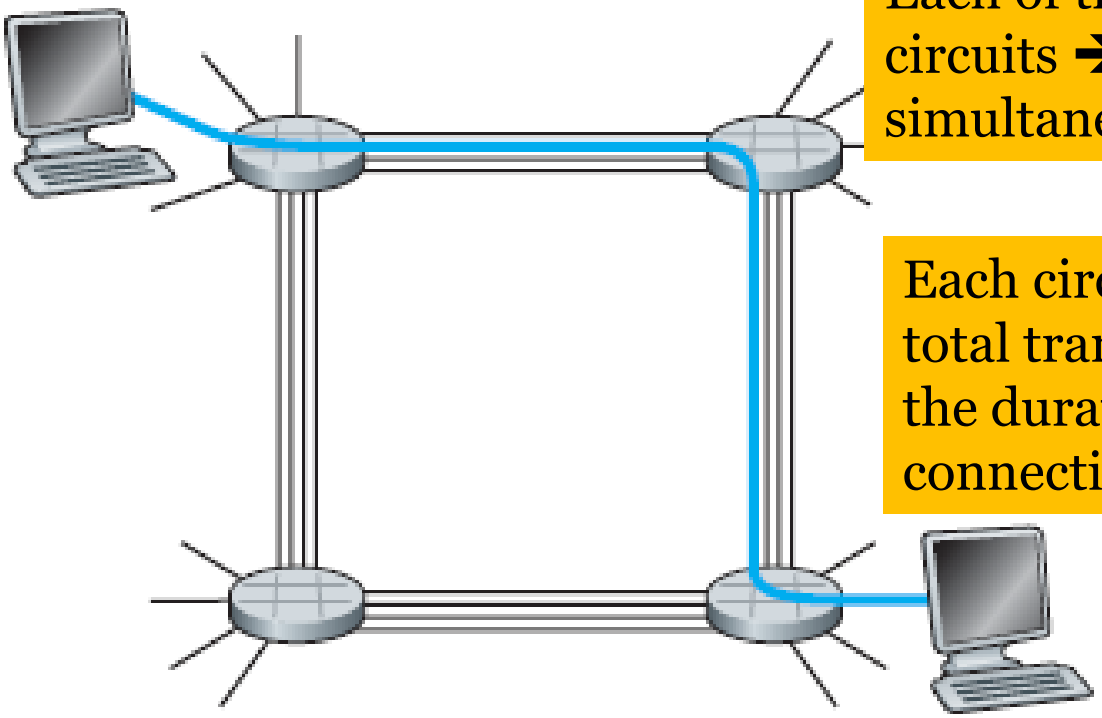
Packet destination: 193.177.1.1

Alternative core: Circuit Switching

- End-to-end resources allocated to, (and reserved for) "call" between source & destination
- In diagram, each link has four circuits.
  - call gets 2nd circuit in top link and 1st circuit in right link.
- Dedicated resources: no sharing
  - circuit-like (guaranteed) performance
- Circuit segment idle if not used by call *(no sharing)*
- Commonly used in traditional telephone networks



The 4 circuit switches are interconnected by 4 links.



Each of these links has four circuits ➔ can support 4 simultaneous connections.

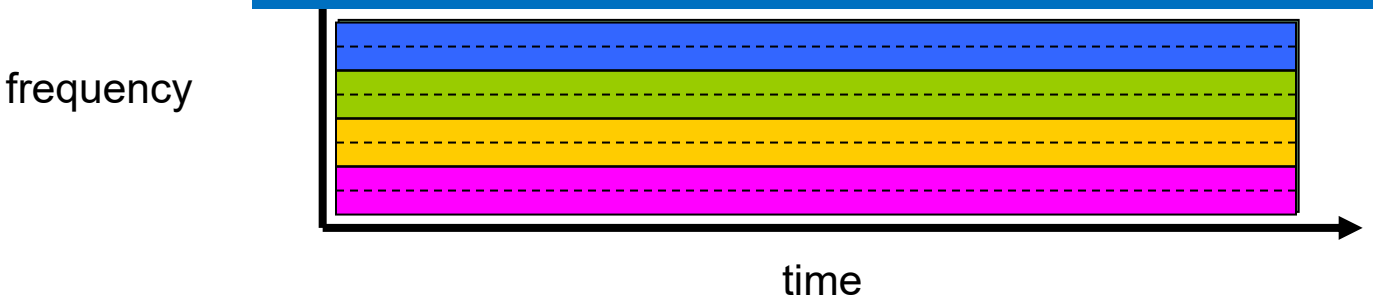
Each circuit ¼ of the link's total transmission capacity for the duration of the connection.

Example:
If a link has a transmission rate of **1 Mbps**
➔ Each connection gets **250 kbps** of dedicated transmission rate.

Circuit switching: FDM versus TDM

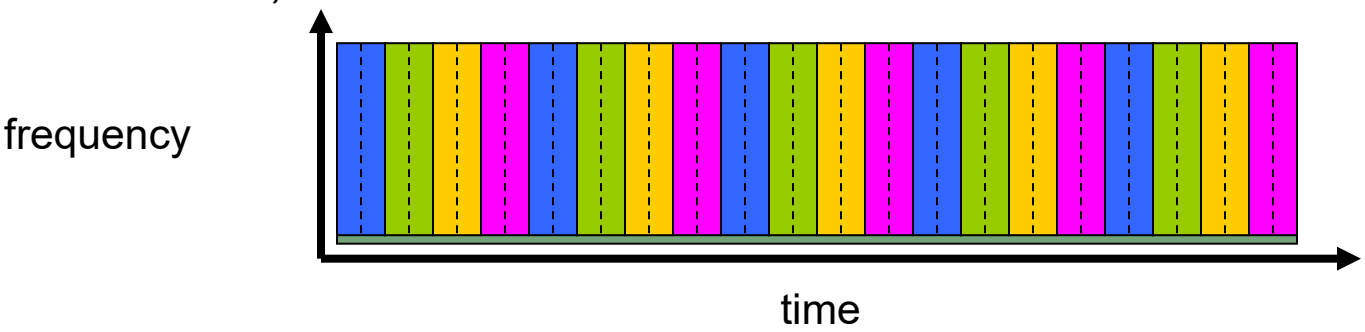
With FDM, each circuit **continuously gets a fraction of the bandwidth.** With TDM, each circuit **gets all of the bandwidth periodically** during brief intervals of time (that is, during slots)

FDM (FREQ DIV MU



Users will access media (e.g. cable) based on DIFFERENT frequency (Hz) allocation

TDM (TIME DIV MUX)



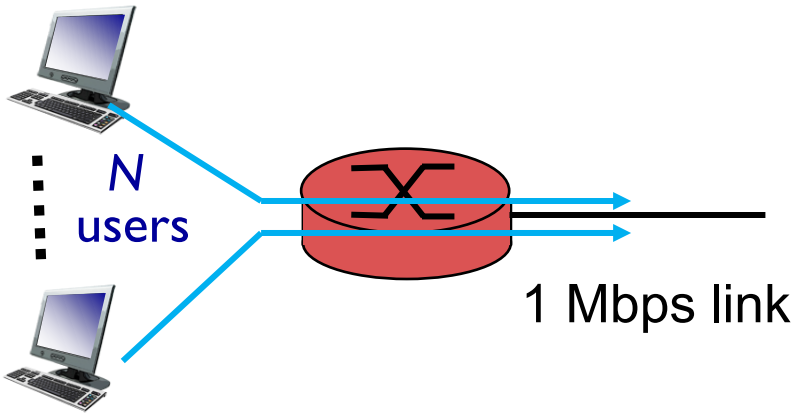
Users will access media (e.g. cable) based on DIFFERENT time slot (second) allocation

Packet switching versus circuit switching

packet switching allows more users to use network!

example:

- 1 Mb/s link
- each user:
  - 100 kb/s when "active"
  - active 10% of time



- *circuit-switching:*
  - 10 users
- *packet switching:*
  - with 35 users, probability > 10 active at same time is less than .0004 *

Packet switching provides essentially the same performance as circuit switching, *but does so while allowing for more than three times the number of users.*

Not part of your syllabus – just extra knowledge

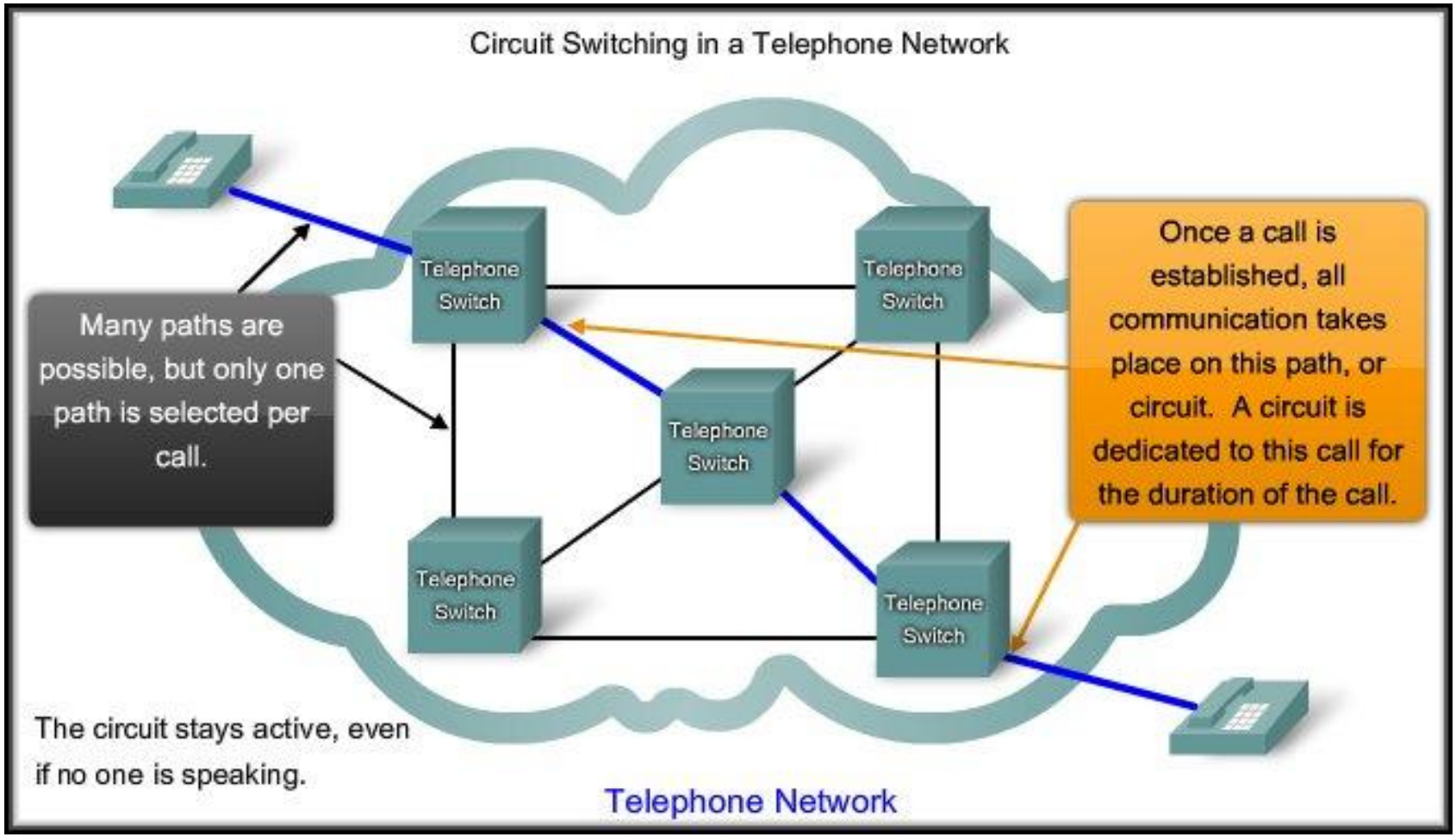
Answer: *Since we can safely assume that each user is active independently from the others, we can model our problem by a binomial distribution* $X \sim B(N,p)$ *where* $N = 35$ *is the total number of users and* $p = 0.1$ *is the probability of being active. The probability* $P(X = k)$ *of having* k *active users simultaneously is defined as* $\binom{N}{k} \cdot p^k \cdot (1-p)^{N-k}$ *where* $\binom{N}{k} = \frac{N!}{k! \cdot (N-k)!}$. *Thus the probability of having more than 10 active users is* $\sum_{i=11}^{35} P(X=i) = 1 - \sum_{i=0}^{10} P(X=i) = 1 - \sum_{i=0}^{10} \frac{35!}{i! \cdot (35-i)!} \cdot p^i \cdot (1-p)^{35-i} \approx 0.0004$

http://mosi.cs.uni-saarland.de/data/DataNetworks13/Homework1-solution.pdf

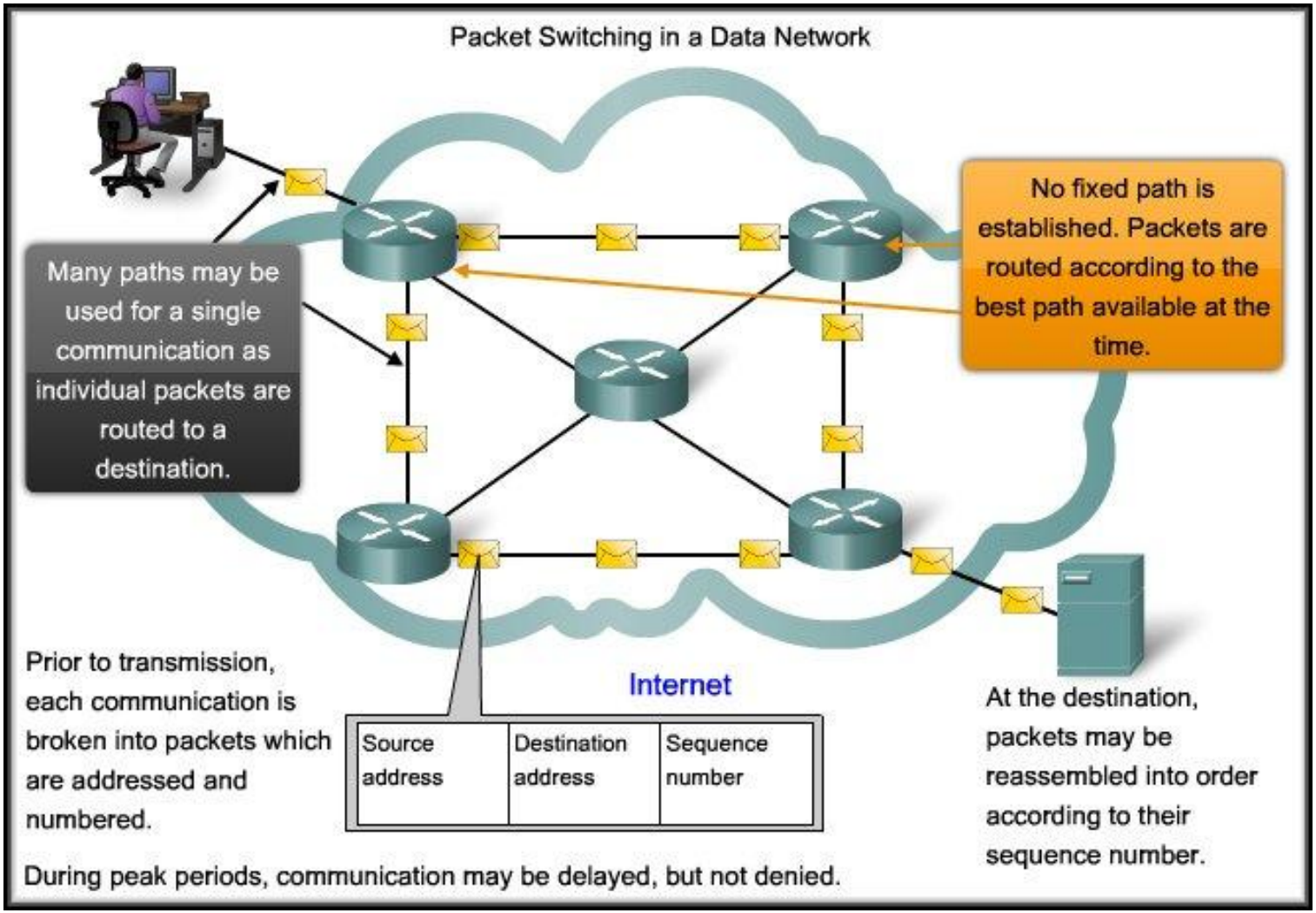
Packet switching versus circuit switching

- Is packet switching a "slam dunk winner?"
- Great for bursty data - sometimes has data to send, but other times not
  - resource sharing
  - simpler, no call setup
- Excessive congestion possible: packet delay and loss due to buffer overflow
  - protocols needed for reliable data transfer, congestion control
- *Q:* How to provide circuit-like behavior with packet switching?
  - bandwidth guarantees needed for audio/video apps
  - still an unsolved problem (chapter 7)

Circuit Switched – Connection-Oriented Networks



Packet Switched – Connectionless Networks



Circuit Switched	Packet Switched
Connection-oriented	Connectionless
Dedicated Circuit	Shared Circuit
Guaranteed level of service (Bandwidth, QoS)	Messages divided into packets
Inefficient use of Medium	Efficient use of Medium
Single path, no redundancy	**Fault Tolerant**, multiple possible paths

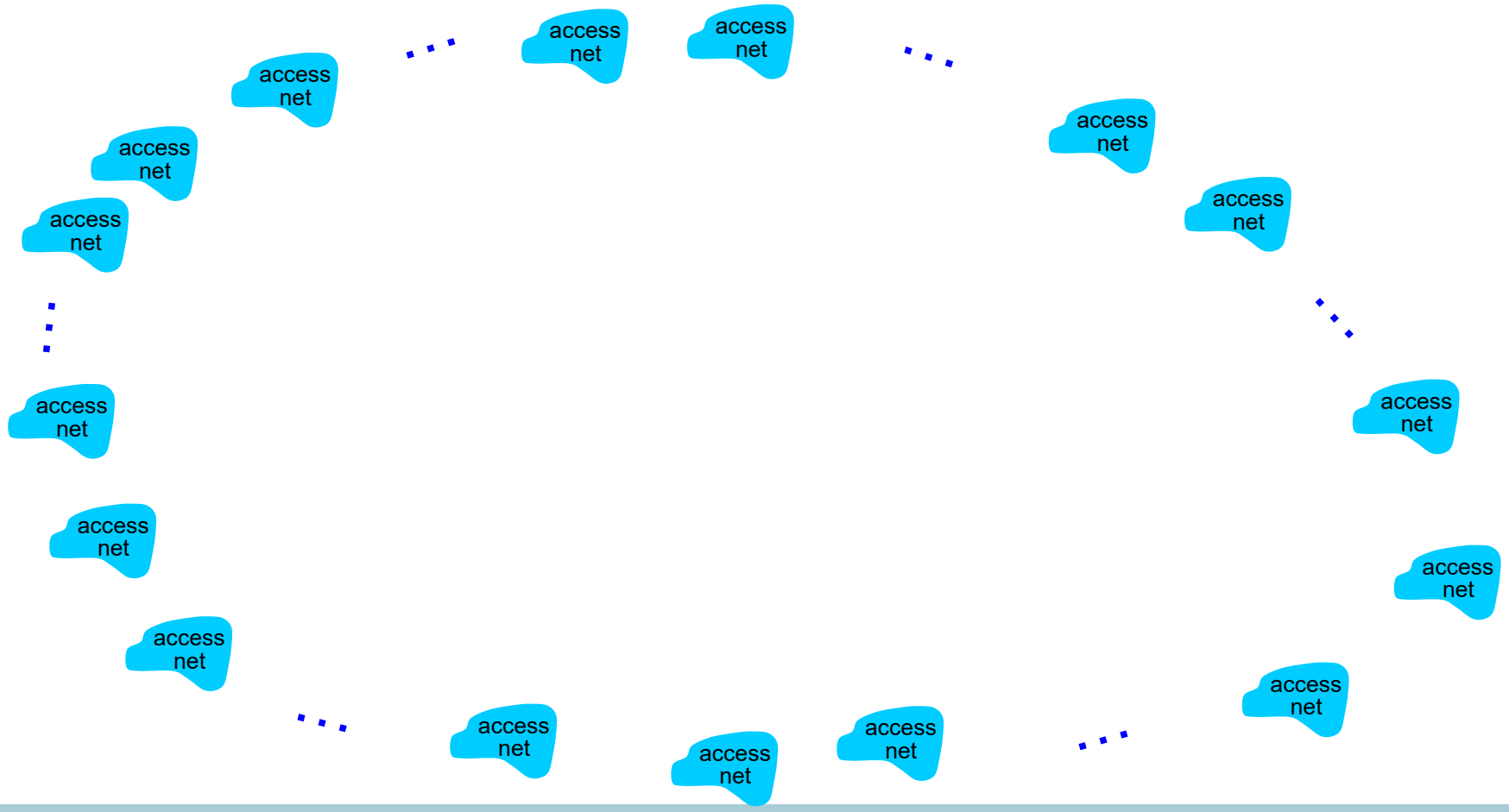
Internet structure: network of networks



- End systems connect to Internet via access ISPs (Internet Service Providers)
  - Residential, company and university ISPs
- Access ISPs in turn must be interconnected.
  - So that any two hosts can send packets to each other
- Resulting network of networks is very complex
  - Evolution was driven by economics and national policies
- Let's take a stepwise approach to describe current Internet structure

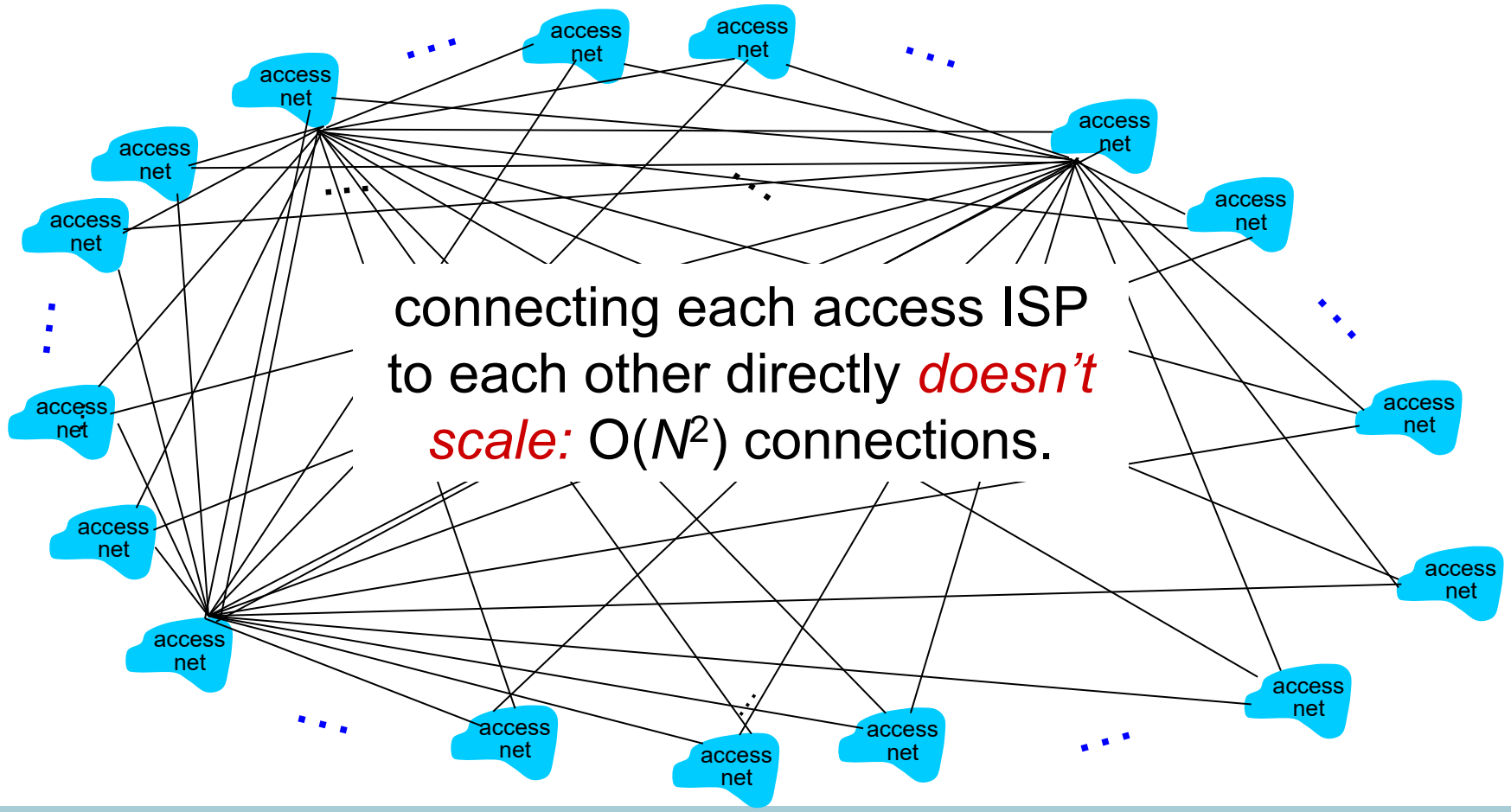
Internet structure: network of networks

Question: given *millions* of access ISPs, how to connect them together?



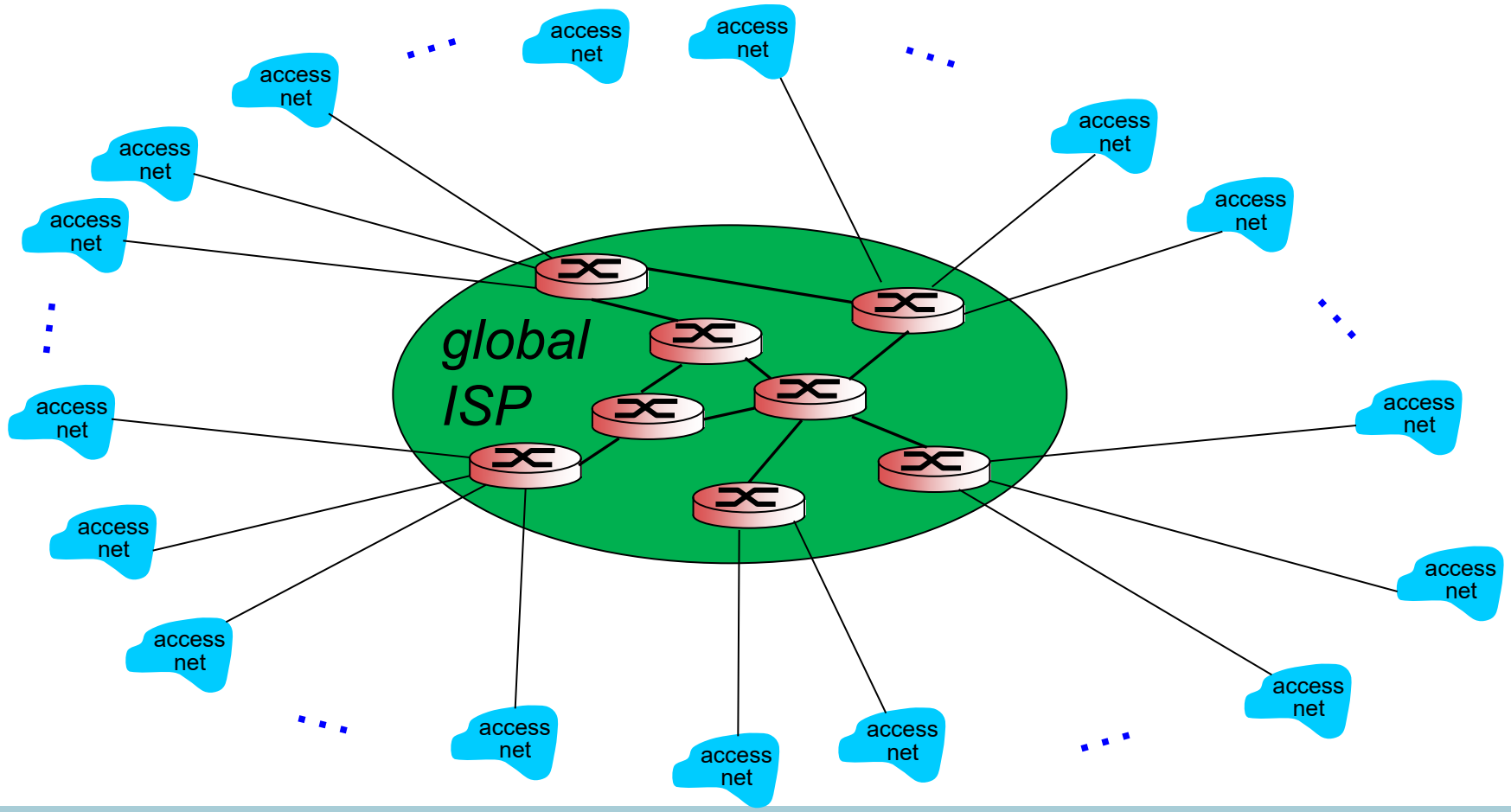
Internet structure: network of networks

Option: connect each access ISP to every other access ISP?



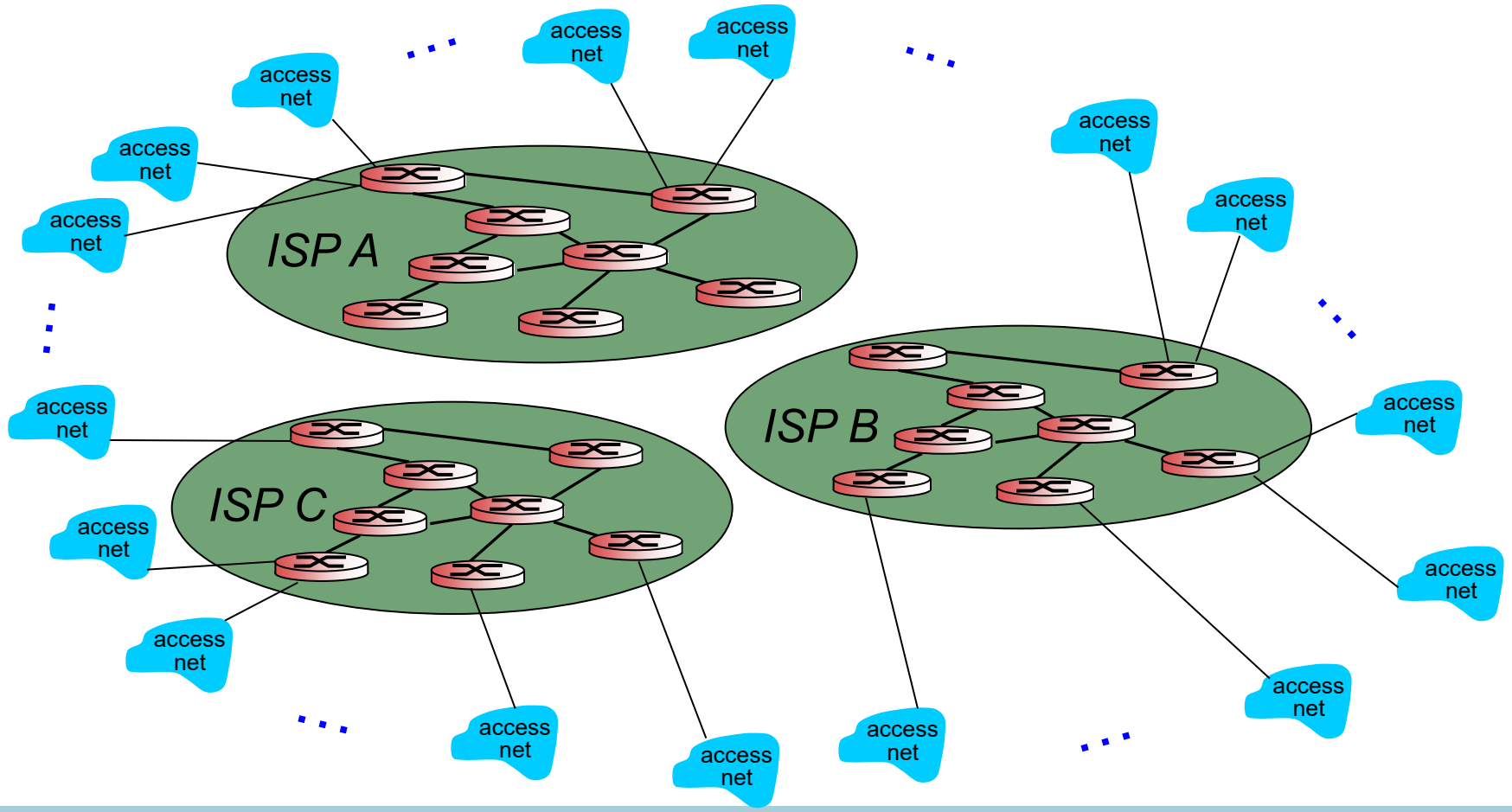
Internet structure: network of networks

Option: connect each access ISP to a global transit ISP? Customer and provider ISPs have economic agreement.



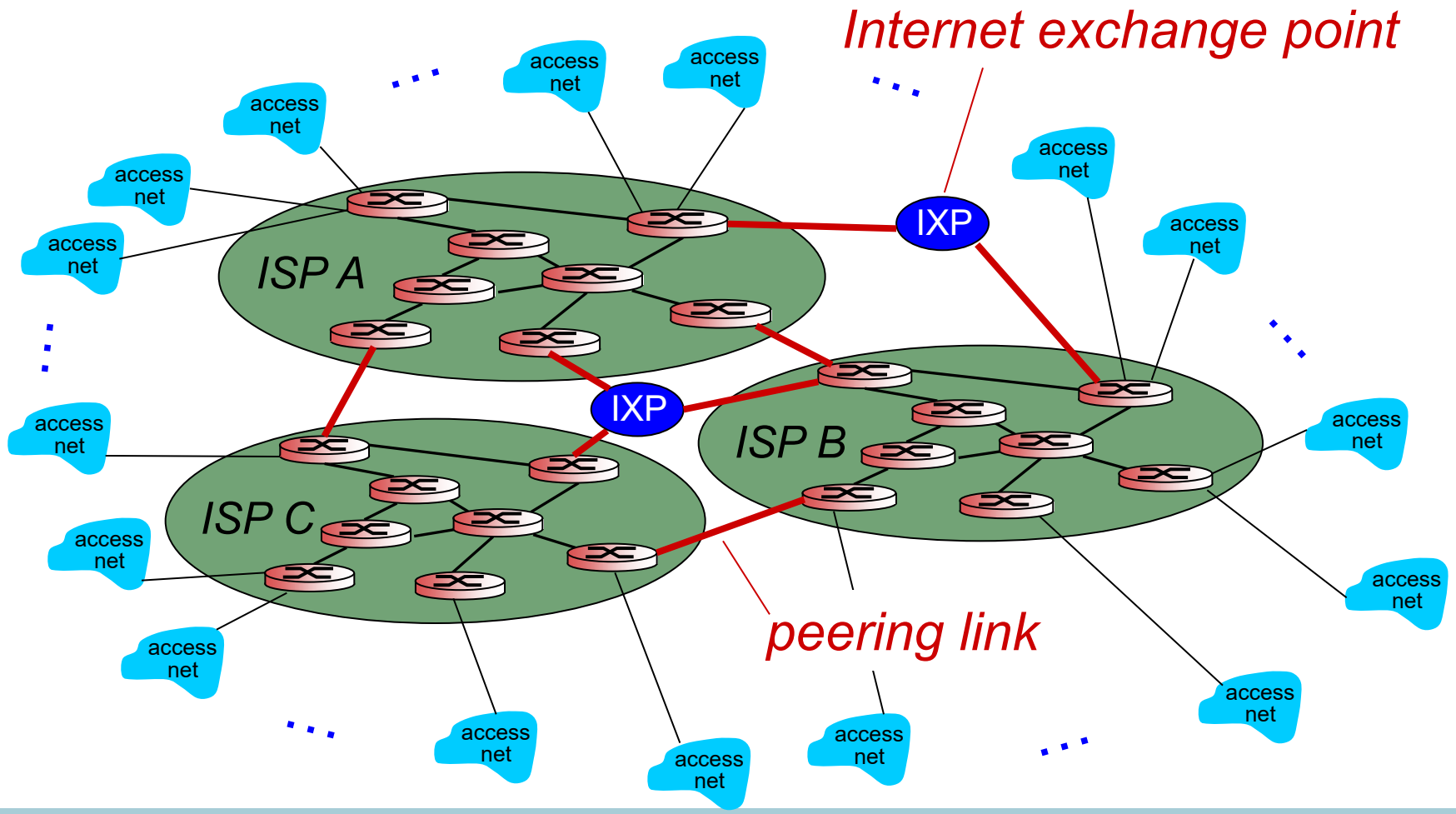
Internet structure: network of networks

But if one global ISP is viable business, there will be competitors ….



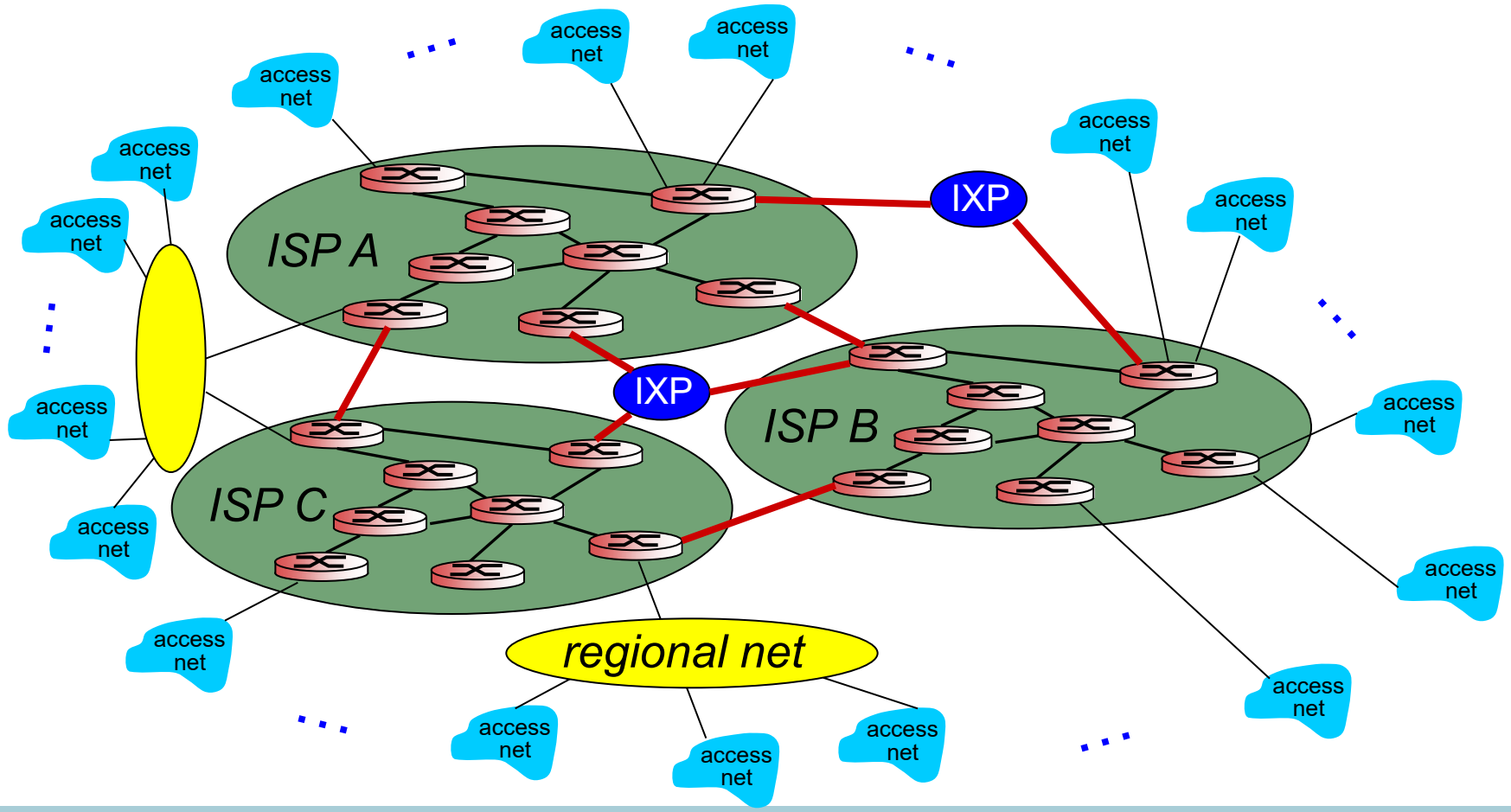
Internet structure: network of networks

But if one global ISP is viable business, there will be competitors …. which must be interconnected



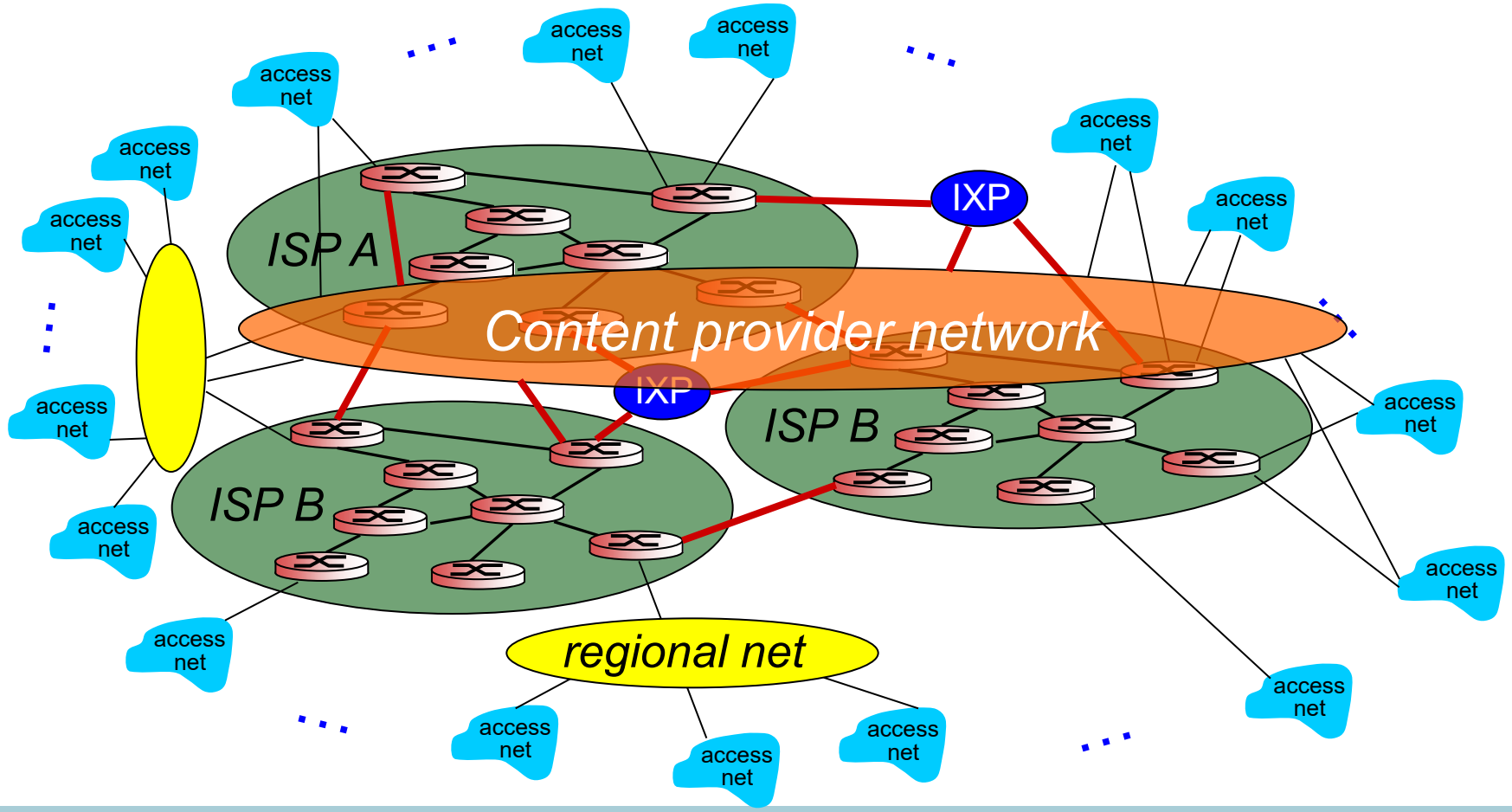
Internet structure: network of networks

… and regional networks may arise to connect access nets to ISPs



Internet structure: network of networks

… and content provider networks (e.g., Google, Microsoft, Akamai) may run their own network, to bring services, content close to end users

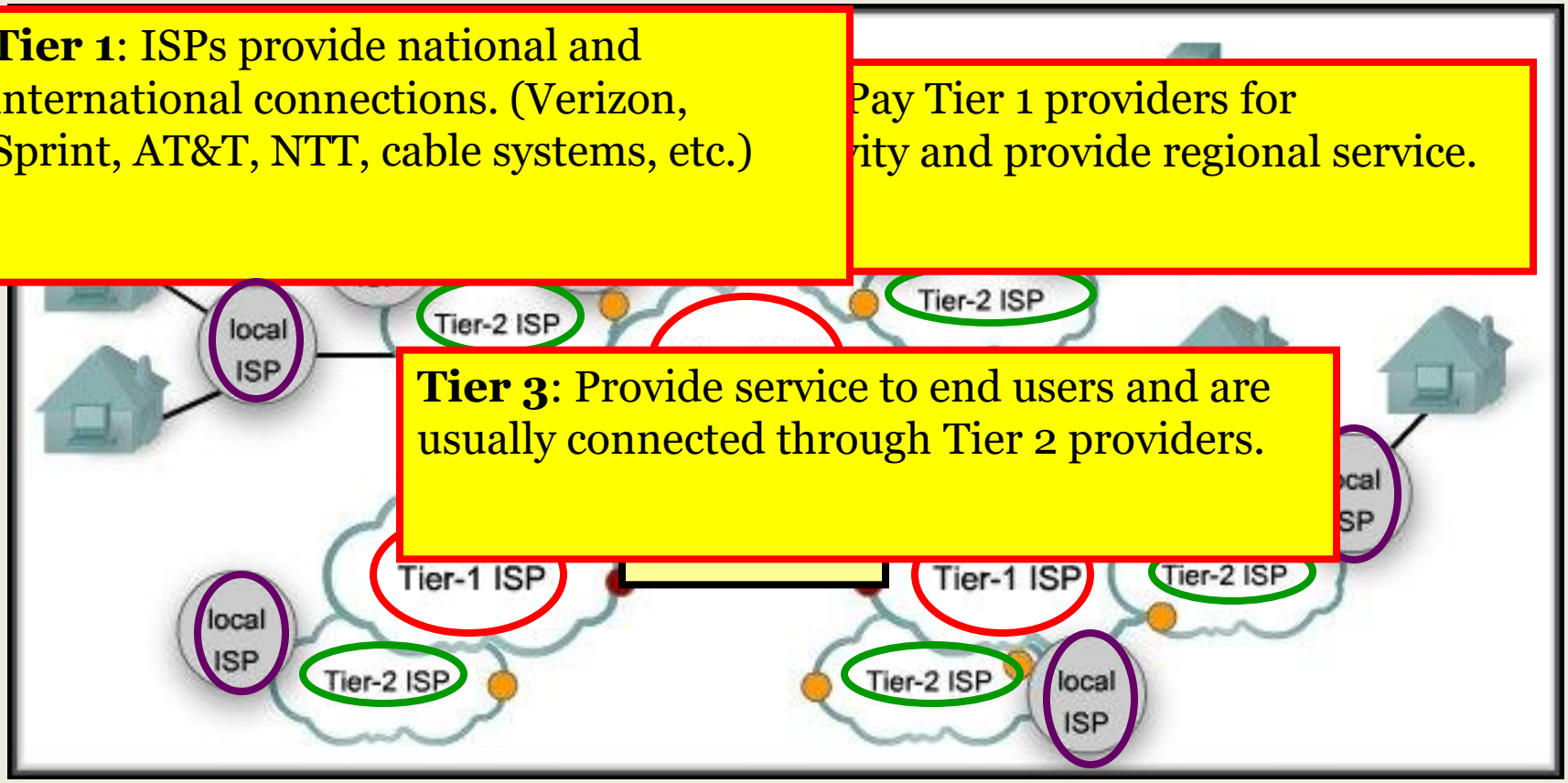


Network Architecture - Scalability

- A good example of scalability is the Tier architecture of the Internet.

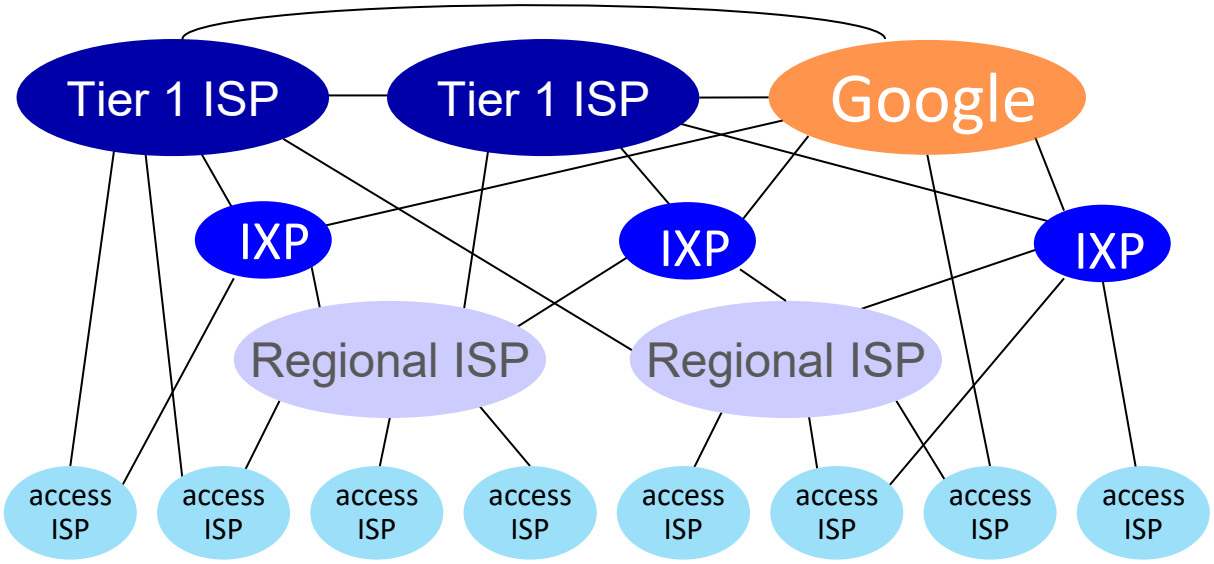
Tier 1: ISPs provide national and international connections. (Verizon, Sprint, AT&T, NTT, cable systems, etc.)

Pay Tier 1 providers for
ity and provide regional service.



Tier 3: Provide service to end users and are usually connected through Tier 2 providers.

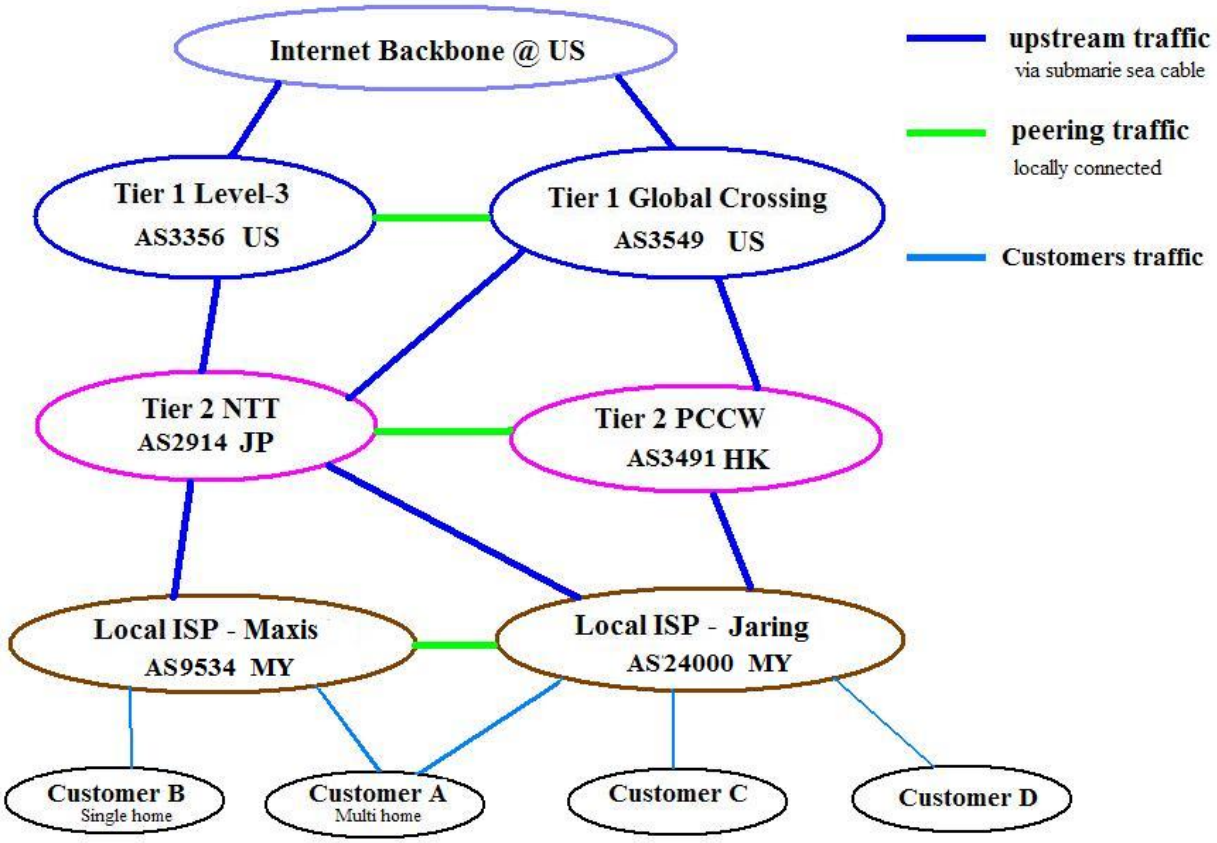
Internet structure: a "network of networks"



At "center": small # of well-connected large networks

- "tier-1" commercial ISPs (e.g., Level 3, Sprint, AT&T, NTT), national & international coverage
- content provider networks (e.g., Google, Facebook): private network that connects its data centers to Internet, often bypassing tier-1, regional ISPs

Internet structure: network of networks



- at center: small number of well-connected large networks
  - "tier-1" commercial ISPs (e.g., Level 3, Sprint, AT&T, NTT), national & international coverage
  - content provider network (e.g, Google): private network that connects it data centers to Internet, often bypassing tier-1, regional ISPs
  - Example ISP list in Malaysia: List , Service Offer , Service Cost

End of class today

Chapter 1: roadmap



1.1 what *is* the Internet?

1.2 network edge

- end systems, access networks, links

1.3 network core

- packet switching, circuit switching, network structure

1.4 delay, loss, throughput in networks

1.5 protocol layers, service models

1.6 networks under attack: security

1.7 history

How do loss and delay occur?

25 people wants to go to Jusco in 5 cars.

From SC,UTM to JUSCO is 50km. Each car's speed is 100km/h. Theoretically → will arrive in 30 minutes

SC,UTM

JUSCO TU

Where are the possible delays?

Arrange who is going

Processing Delay

Get into cars

Queuing Delay

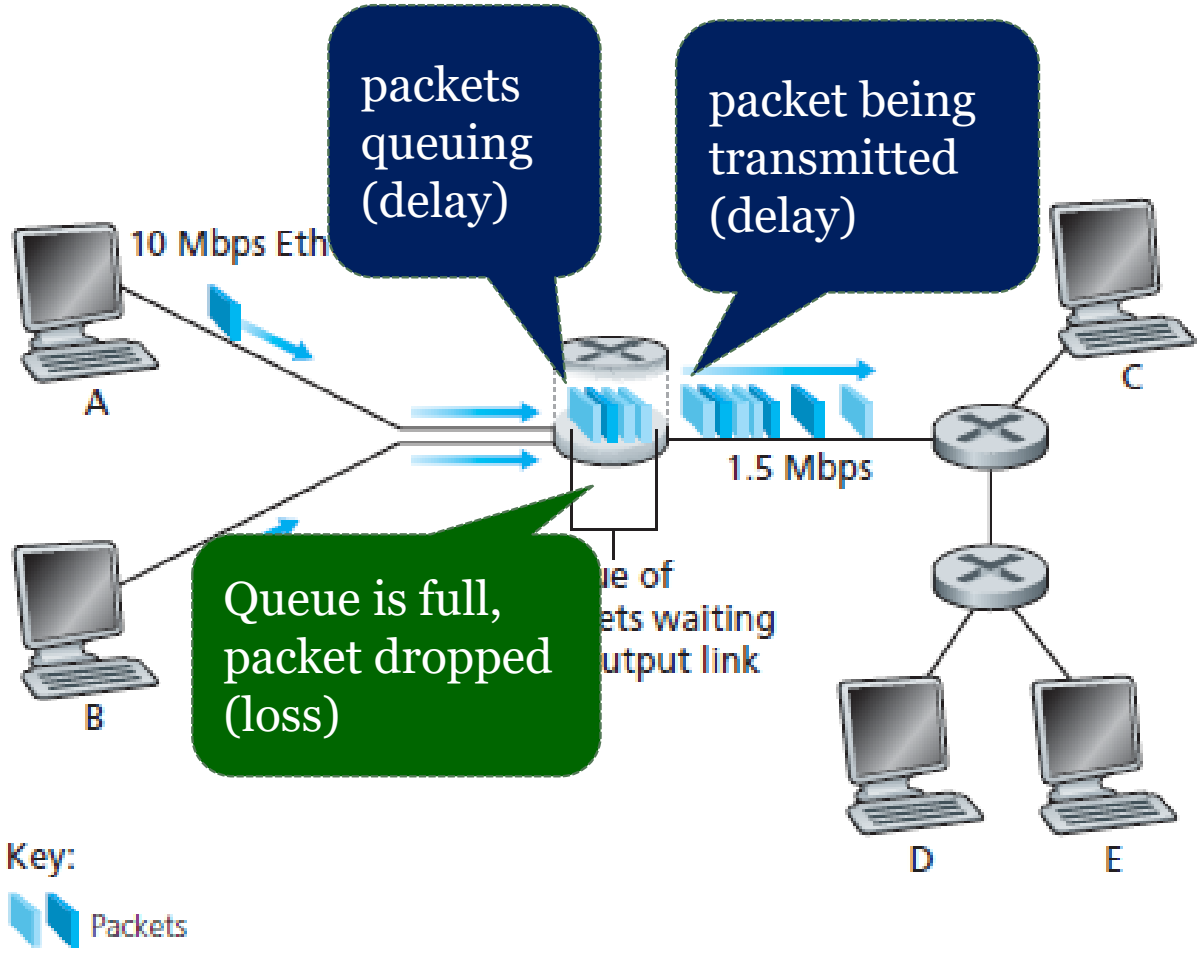
Cars wait to move

Transmission Delay

Condition of road/traffic

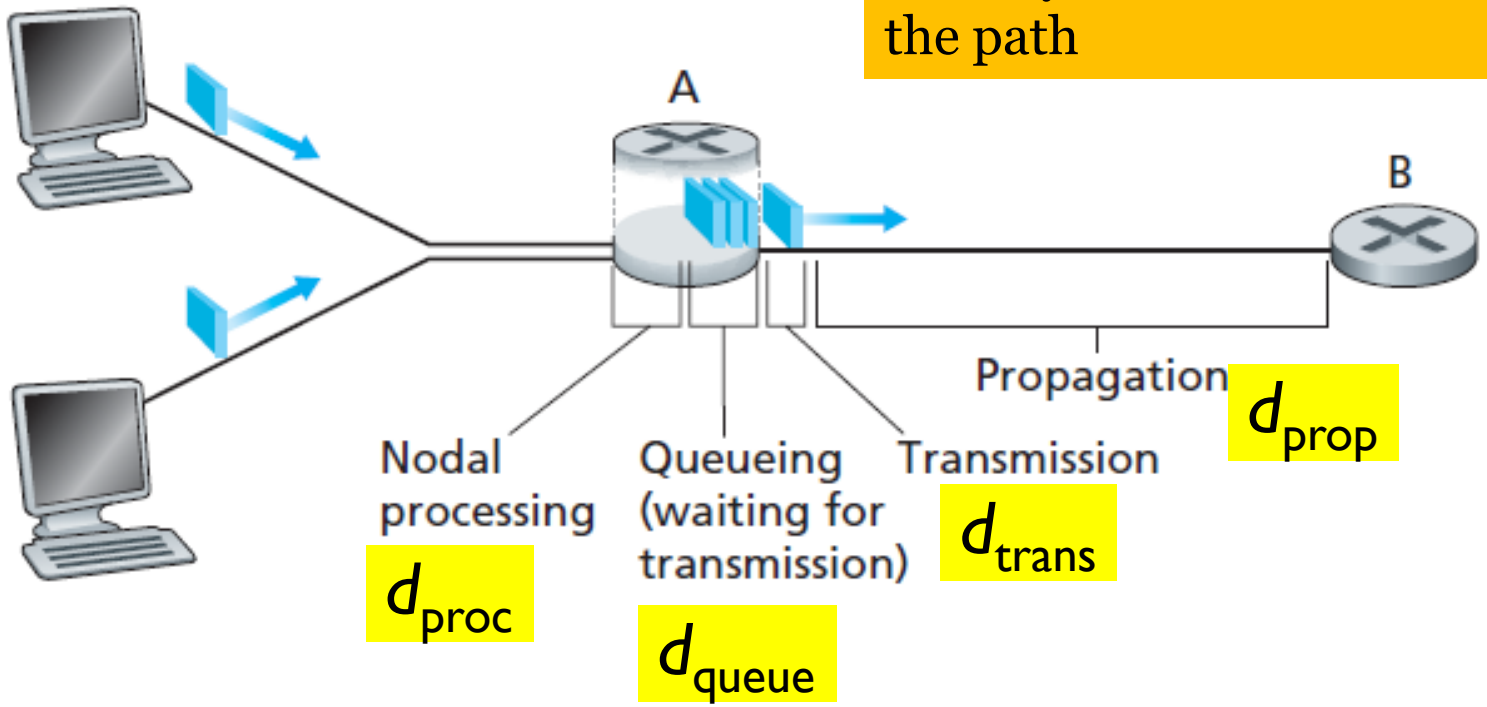
Propagation Delay

How do loss and delay occur?



The nodal delay at router A

A packet goes through a series of delays at **each node** along the path



$$\textit{Total nodal delay} = d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

d_{proc}: nodal processing

- Examine packet header
- Determine next path
- check bit errors
- determine output link
- typically < microsecs

d_{queue}: queueing delay

- time waiting at output link for transmission
- depends on congestion level of router

d_{trans}: transmission delay:

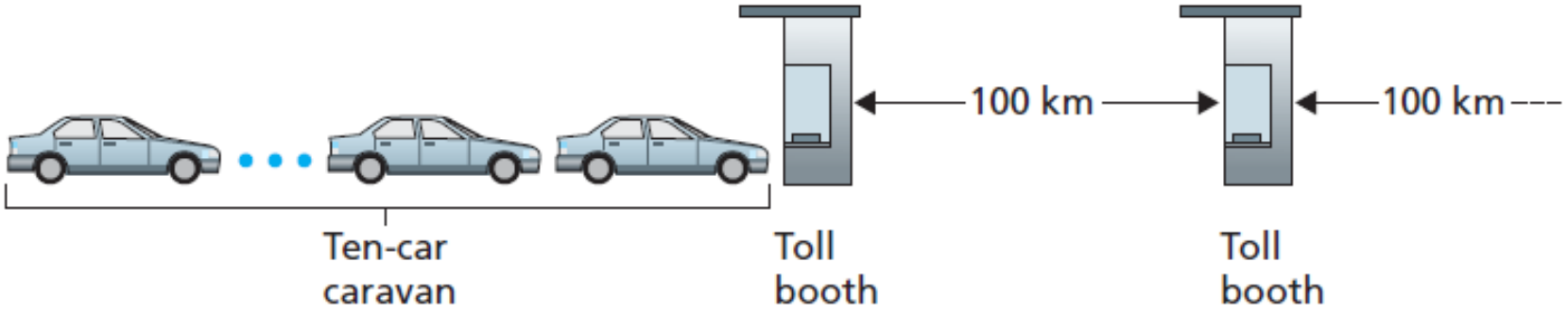
- *L*: packet length (bits)
- *R*: link *bandwidth (bps)*
- $d_{trans} = L/R$
- the amount of time required to push (that is, transmit) all of the packet's bits into the link.

d_{prop}: propagation delay:

- Propagation speed depends on the physical medium of the link (range $2x10^8$ to $3x10^8$ meters/sec)
- *d*: length of physical link
- *s*: propagation speed in medium
- $d_{prop} = d/s$

d_{trans} and d_{prop} *very* different

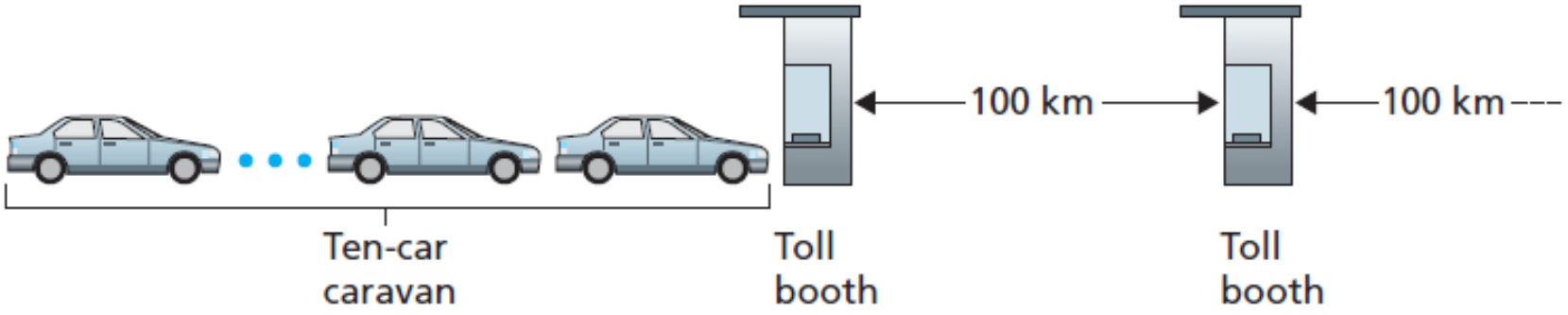
Caravan analogy



- cars "propagate" at 100 km/hr
- toll booth takes 12 sec to service car (bit transmission time)
- car~bit; caravan ~ packet
- *Q:* How long until caravan is lined up before 2nd toll booth?

- time to "push" entire caravan through toll booth onto highway = 12*10 = 120 sec
- time for last car to propagate from 1st to 2nd toll both: 100km/(100km/hr)= 1 hr
- *A:* 62 minutes

Caravan analogy



- suppose cars now "propagate" at 1000 km/hr
- and suppose toll booth now takes one min to service a car

- *Q:* Will cars arrive to 2nd booth before all cars serviced at first booth?

- *A: Yes!* after 7 min, 1st car arrives at second booth; three cars still at 1st booth.

Given the following conditions:

- **Speed of cars** = 1000 km/hr.
- **Service time per car** = 1 minute.
- **Distance between toll booths** = 100 km.
- **Number of cars** = 10.

Step 1: Calculate the propagation time (time for the first car to reach the second booth)

First, convert the speed into km/min:

$$\text{Speed in km/min} = \frac{1000\,\text{km/hr}}{60\,\text{min/hr}} = 16.67\,\text{km/min}$$

Now, calculate the time it takes for the first car to travel 100 km:

$$\text{Propagation Time} = \frac{100\,\text{km}}{16.67\,\text{km/min}} \approx 6\,\text{minutes}$$

Step 2: Calculate the total service time at the first toll booth

Each car takes 1 minute to be serviced. For 10 cars, the total service time is:

$$\text{Total Service Time} = 10 \times 1\,\text{minute} = 10\,\text{minutes}$$

Step 3: Compare propagation time and service time

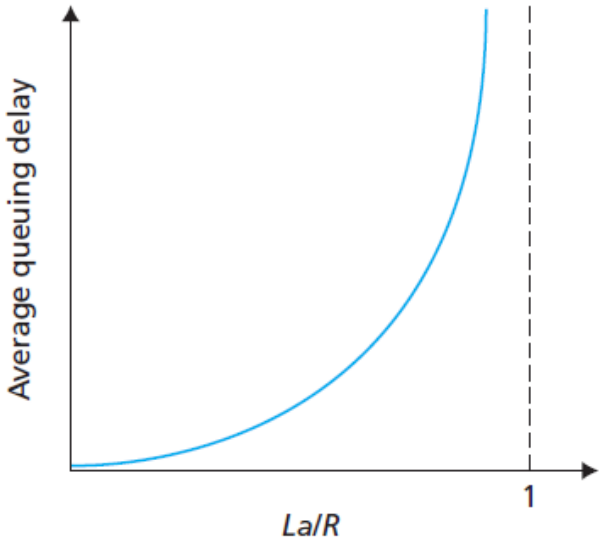
The first car will reach the second toll booth in **6 minutes**, and it takes **10 minutes** to service all 10 cars at the first toll booth.

- Since **6 minutes < 10 minutes**, the first car **will arrive at the second toll booth before all 10 cars are serviced** at the first booth.

With a distance of 100 km between the toll booths, the first car will arrive at the second booth in **6 minutes**, while the last car will still be in service at the first toll booth (since the total service time for all 10 cars is 10 minutes). Therefore, the first car will reach the second booth before all cars are serviced at the first booth.

Queuing delay (revisited)

- *R:* link bandwidth (bps)
- *L:* packet length (bits)
- a: average packet arrival rate (a is in units of packets/sec).



traffic intensity = La/R

- *La/R* ~ 0: avg. queueing delay small
- La/R ≤ 1 : avg. queueing delay large
- *La/R > 1* : more "work" arriving than can be serviced, average delay infinite!



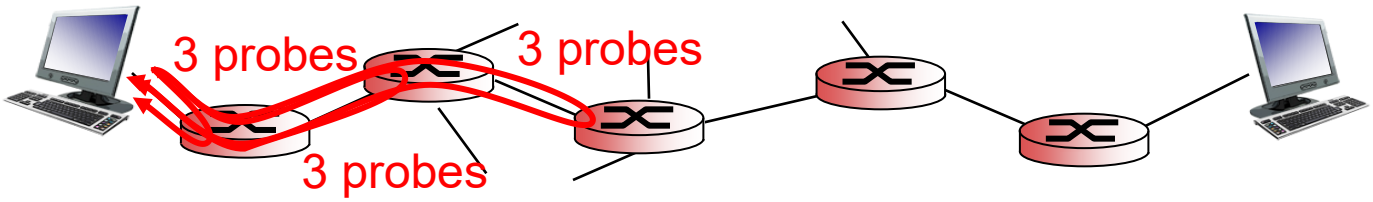
La/R ~ 0



La/R ≤ 1

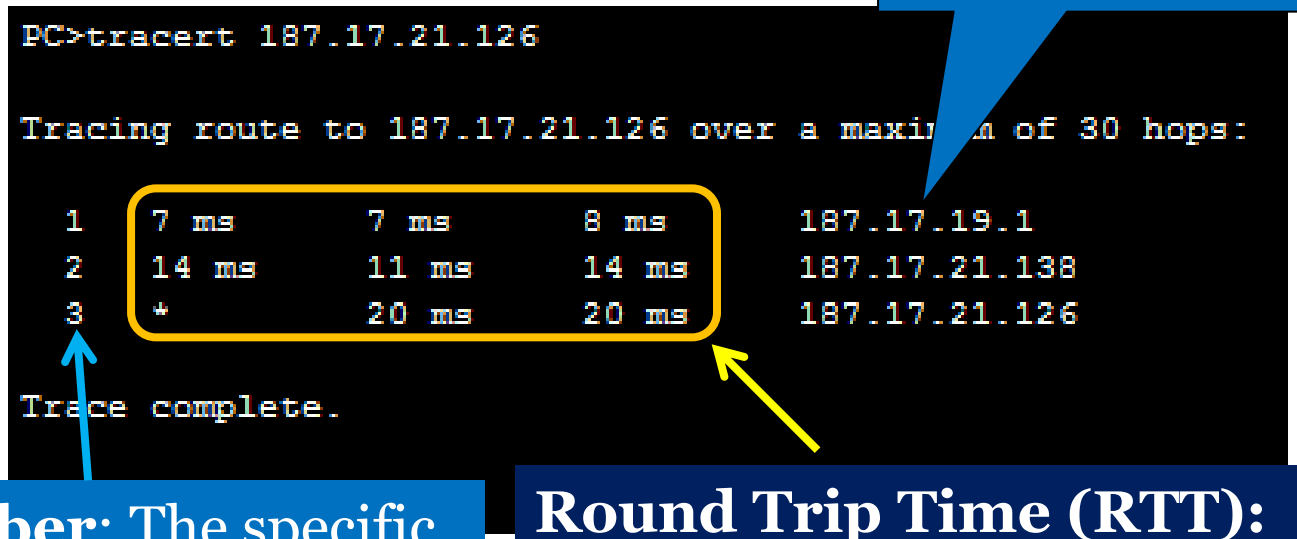
"Real" Internet delays and routes

- What do "real" Internet delay & loss look like?
- `traceroute` program: provides delay measurement from source to router along end-end Internet path towards destination. For all *i*:
  - sends three packets that will reach router *i* on path towards destination
  - router *i* will return packets to sender
  - sender times interval between transmission and reply.



the destination system

The IP address of the hop



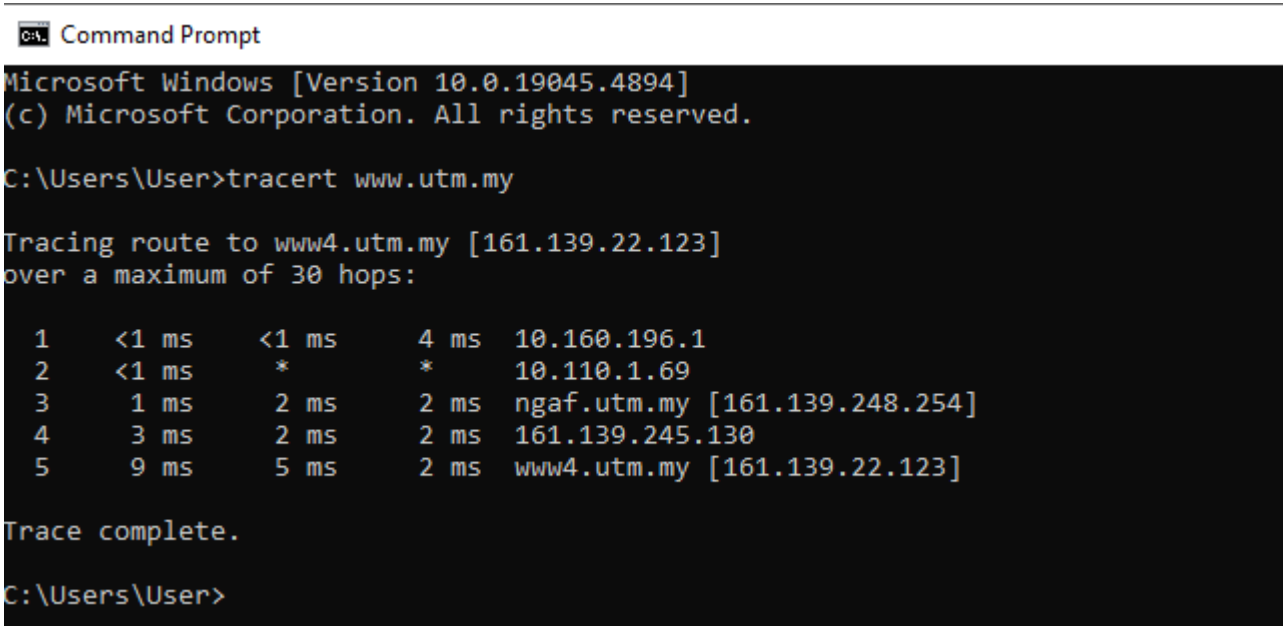
Hop number: The specific hop number in the path from the sender to the destination.

Round Trip Time (RTT):
The time it takes for a packet to get to a hop and back, displayed in milliseconds (ms).
➔ By default, **tracert** sends 3 packets to each hop, so the output lists three roundtrip times per hop.

Lets try

- Open Command prompt
  - Go the search and type cmd
  - Type tracert www.utm.my





"Real" Internet delays, routes

traceroute: gaia.cs.umass.edu to www.eurecom.fr



3 delay measurements from gaia.cs.umass.edu to cs-gw.cs.umass.edu

```
1  cs-gw (128.119.240.254)  1 ms  1 ms  2 ms
2  border1-rt-fa5-1-0.gw.umass.edu (128.119.3.145)  1 ms  1 ms  2 ms
3  cht-vbns.gw.umass.edu (128.119.3.130)  6 ms 5 ms 5 ms
4  jn1-at1-0-0-19.wor.vbns.net (204.147.132.129)  16 ms 11 ms 13 ms
5  jn1-so7-0-0-0.wae.vbns.net (204.147.136.136)  21 ms 18 ms 18 ms
6  abilene-vbns.abilene.ucaid.edu (198.32.11.9)  22 ms  18 ms  22 ms
7  nycm-wash.abilene.ucaid.edu (198.32.8.46)  22 ms  22 ms  22 ms
8  62.40.103.253 (62.40.103.253)  104 ms 109 ms 106 ms
9  de2-1.de1.de.geant.net (62.40.96.129)  109 ms 102 ms 104 ms
10  de.fr1.fr.geant.net (62.40.96.50)  113 ms 121 ms 114 ms
11  renater-gw.fr1.fr.geant.net (62.40.103.54)  112 ms  114 ms  112 ms
12  nio-n2.cssi.renater.fr (193.51.206.13)  111 ms  114 ms  116 ms
13  nice.cssi.renater.fr (195.220.98.102)  123 ms  125 ms  124 ms
14  r3t2-nice.cssi.renater.fr (195.220.98.110)  126 ms  126 ms  124 ms
15  eurecom-valbonne.r3t2.ft.net (193.48.50.54)  135 ms  128 ms  133 ms
16  194.214.211.25 (194.214.211.25)  126 ms  128 ms  126 ms
17  * * *
18  * * *
19  fantasia.eurecom.fr (193.55.113.142)  132 ms  128 ms  136 ms
```

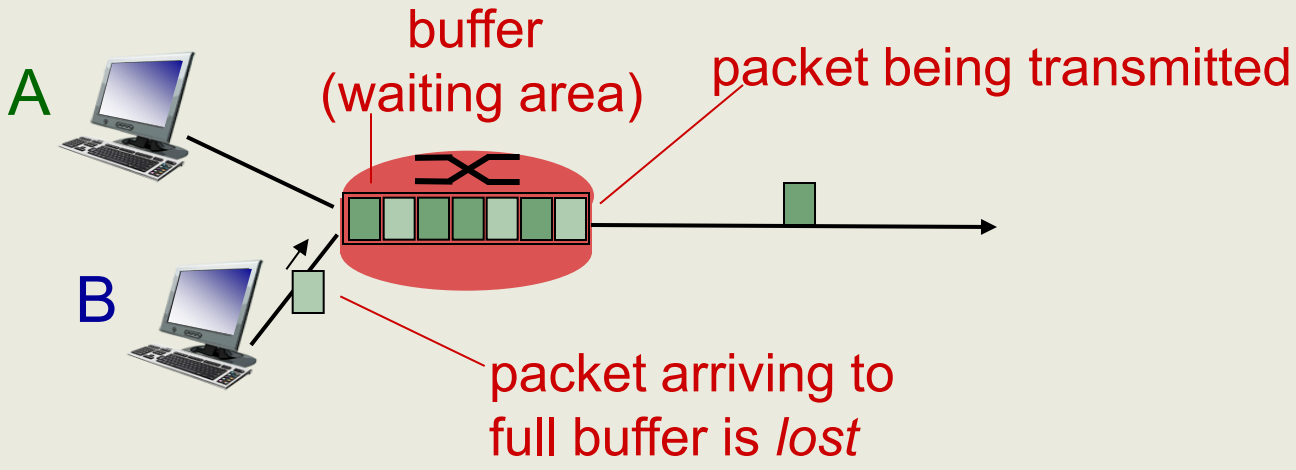
trans-oceanic link

* means no response (probe lost, router not replying)

* Do some traceroutes at www.traceroute.org

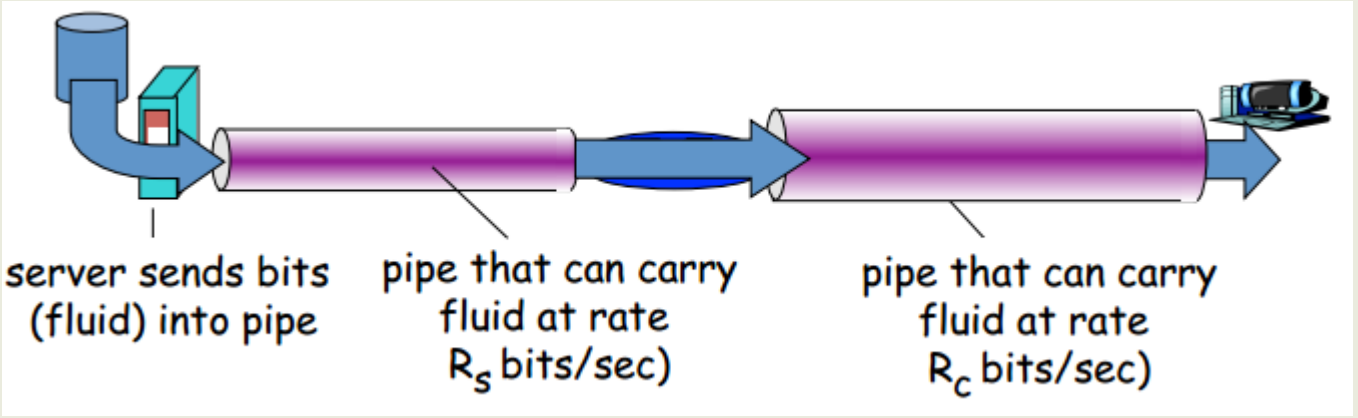
Packet loss

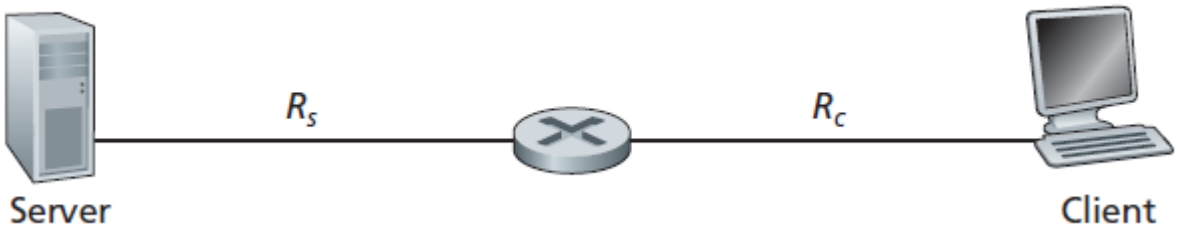
- Queue (aka buffer) has finite capacity
- Packet arriving to a full queue dropped (aka lost)
- Lost packet may be retransmitted by previous node, by source end system, or not at all



Throughput in Computer Networks

- A critical performance measure in computer networks
- Throughput: rate (bits/time unit) at which bits transferred between sender-receiver
  - instantaneous: rate at given point in time
  - average: rate over longer period of time



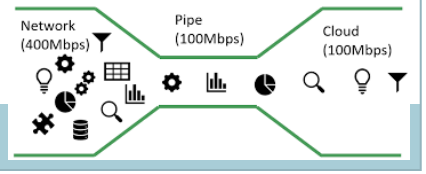


- If $R_s < R_c$
  - Flow from Server to Client not interrupted
  - Throughput = R_s

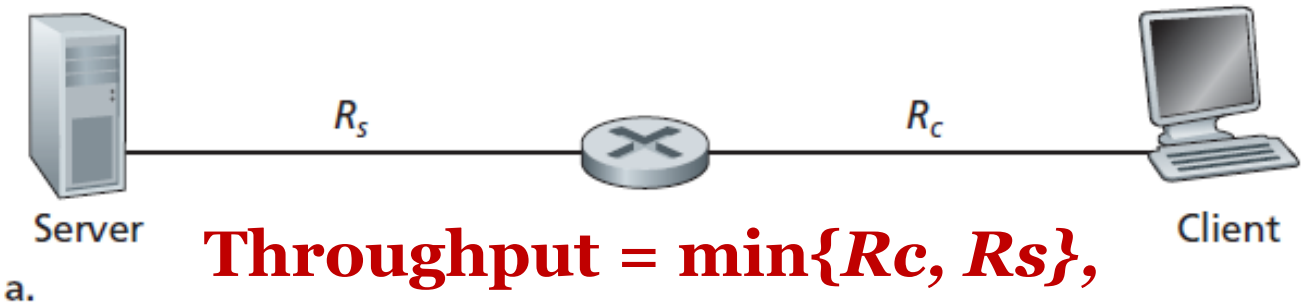
- If $R_s > R_c$
  - Flow from Server to Client is interrupted, queue may form
  - Throughput = R_c

Throughput = min{*Rc, Rs*},

➔ the transmission rate of the **bottleneck link**

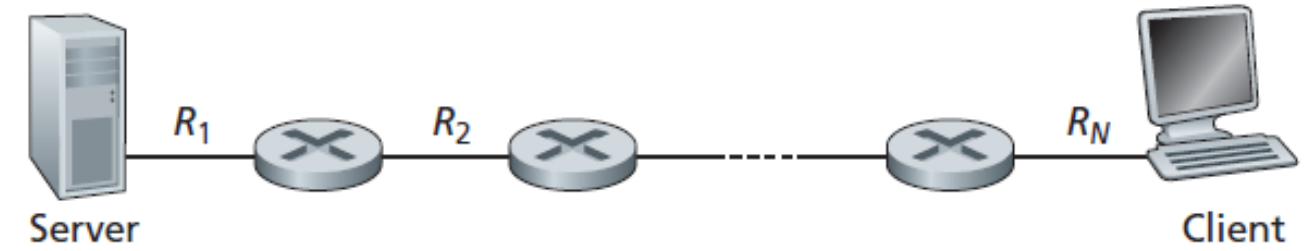


Throughput for a file transfer from server to client



a.

Throughput = min{*Rc, Rs*},



b.

Throughput = min{*R1, R2,…, RN*},

Example A

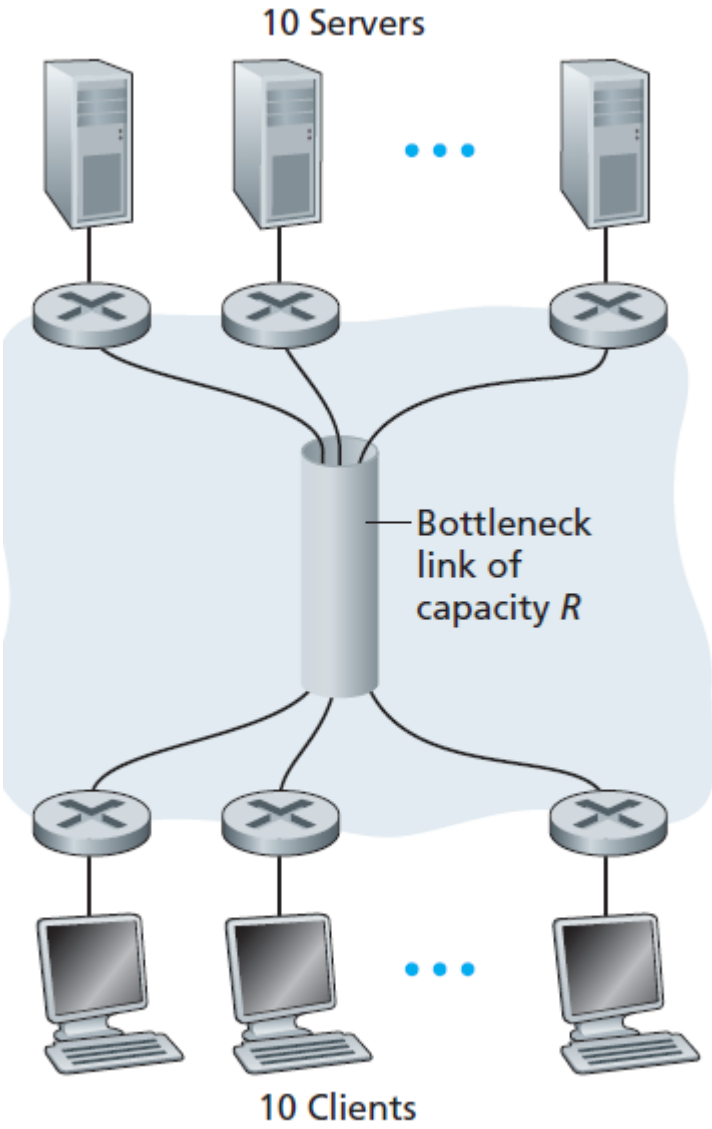
- The time it takes to transfer a large file of *F bits from server to* client as *F/min{Rs, Rc}*
- *Say, you are downloading a* file of *F = 32 million bits, the server has a transmission rate of Rs = 2 Mbps,* and you have an access link of *Rc = 1 Mbps.*
- *The time needed to transfer the file is*

Transfer time = (32 x 10^6) bits/1Mbps
= 32 seconds

**Note: Of course, these expressions for throughput and transfer time are only approximations ➔ as they do not account for store-and-forward and processing delays as well as protocol issues.

End-to-end throughput: Internet scenario

- There are 10 simultaneous downloads taking place, involving 10 client-server pairs.
- The 10 connections (fairly) share backbone bottleneck link R bits/sec
- In practice: R_c or R_s is often bottleneck

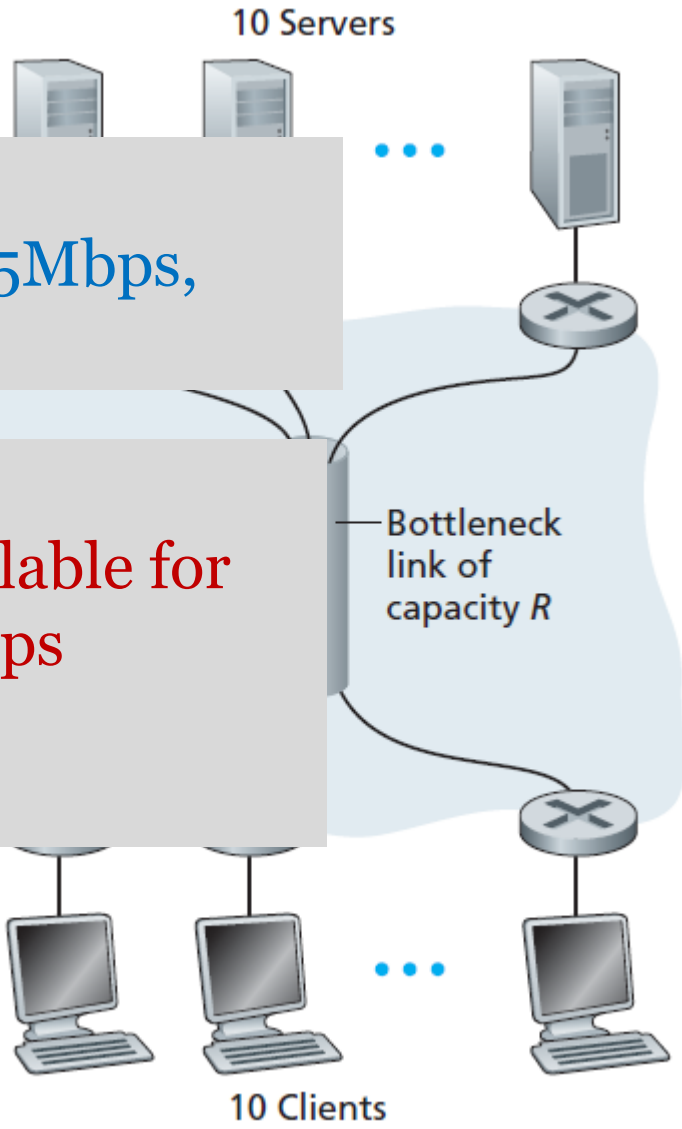


End-to-end throughput: Internet scenario

Example B:
If R_s = 2Mbps, R_c = 1Mbps, R = 5Mbps, what is the throughput?

Answer:
Because R is shared, what is available for each link is (5Mbps/10) = 500kbps

Throughput = 500kbps



Example C

Suppose Host A wants to send a large file to Host B. The path from Host A to Host B has 3-links, of rate R_1=150kbps, R_2=2Mbps, and R_3=1Mbps.

i) Assuming no other traffic in the network, what is the throughput for the file transfer? [2 marks]

Throughput = min{*R1, R2, R3*} = *150kbps*

ii) Suppose the file is 4 million bytes. Roughly, how long will it take to transfer the file to Host B? [5 marks]

Time to transfer= $4*10^6*8/(150*10^3)$=213.3 seconds

Chapter 1: roadmap



1.1 what *is* the Internet?

1.2 network edge

- end systems, access networks, links

1.3 network core

- packet switching, circuit switching, network structure

1.4 delay, loss, throughput in networks

1.5 protocol layers, service models

1.6 networks under attack: security

1.7 history

Protocol "layers"

Networks are complex, with many "pieces":

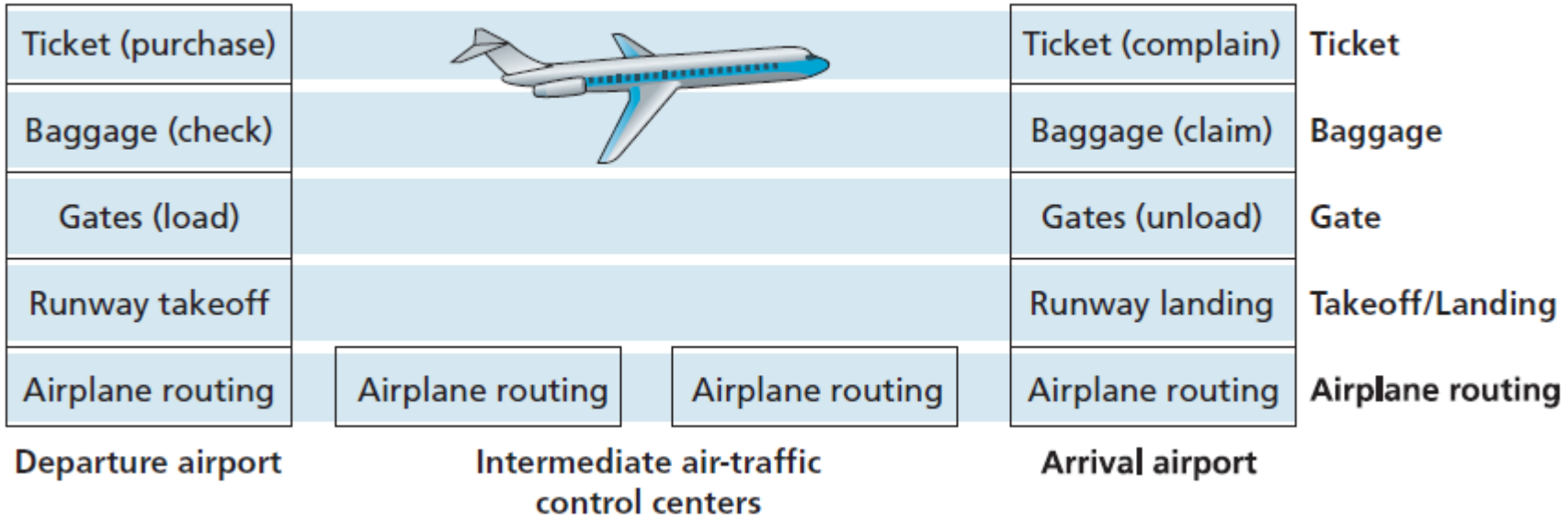
- hosts
- routers
- links of various media
- applications
- protocols
- hardware, software

Question:

is there any hope of *organizing* structure of network?

…. or at least our discussion of networks?

Layered Architecture: Airline functionality



- ***layers:*** each layer implements a service
  - via its own internal-layer actions
  - relying on services provided by layer below

It is a series of steps

Why layering?

Dealing with complex systems:

- Explicit structure allows identification, relationship of complex system' s pieces
  - layered *reference model* for discussion
- Modularization eases maintenance, updating of system
  - change of implementation of layer' s service transparent to rest of system
  - e.g., change in gate procedure doesn't affect rest of system

ISO/OSI reference model

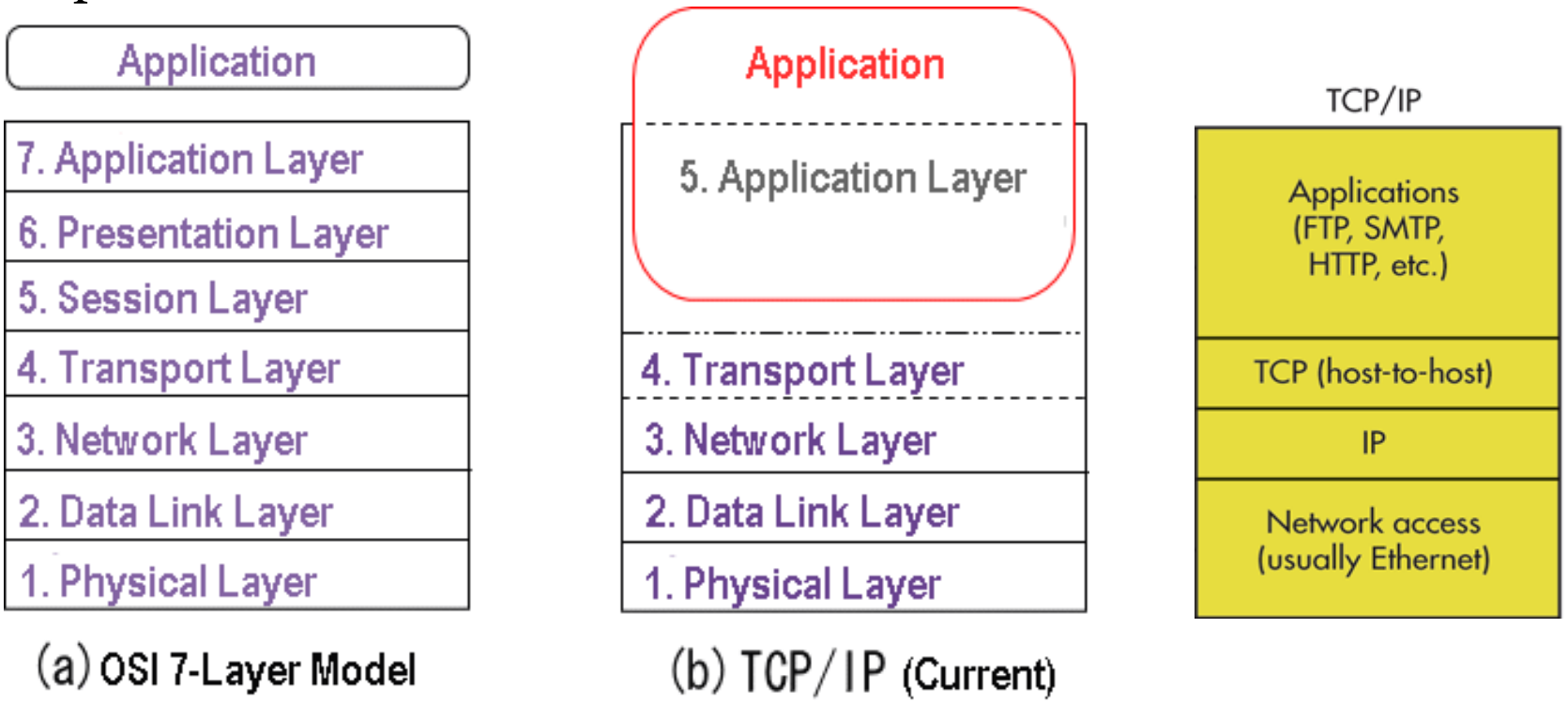
***O**pen **S**ystem **I**nterconnection (OSI)* model defines a **generic** networking framework to implement protocols in seven (7) layers

Generic OSI Model

	Data unit	Layer	Function
Host layers	Data	7. Application	Network process to application
		6. Presentation	allow applications to interpret meaning of data, e.g. data representation, encryption and decryption, convert machine dependent data to machine independent data
		5. Session	Interhost communication, managing sessions between applications, synchronization, data recovery
	Segments	4. Transport	Reliable delivery of packets between points on a network.
Media layers	Packet/Datagram	3. Network	Addressing, routing and (not necessarily reliable) delivery of datagrams between points on a network.
	Bit/Frame	2. Data link	A reliable direct point-to-point data connection.
	Bit	1. Physical	A (not necessarily reliable) direct point-to-point data connection.

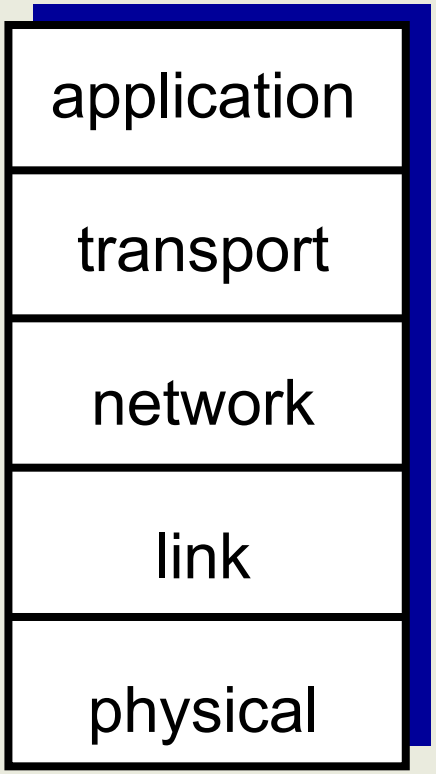
Internet protocol stack (TCP/IP model)

TCP/IP was developed by Advanced Research Projects Agency (ARPA) to build a nationwide packet data network in 1960s. It was first used in UNIX-based computers in universities and government installations. Today, it is the main protocol used in all Internet operations.



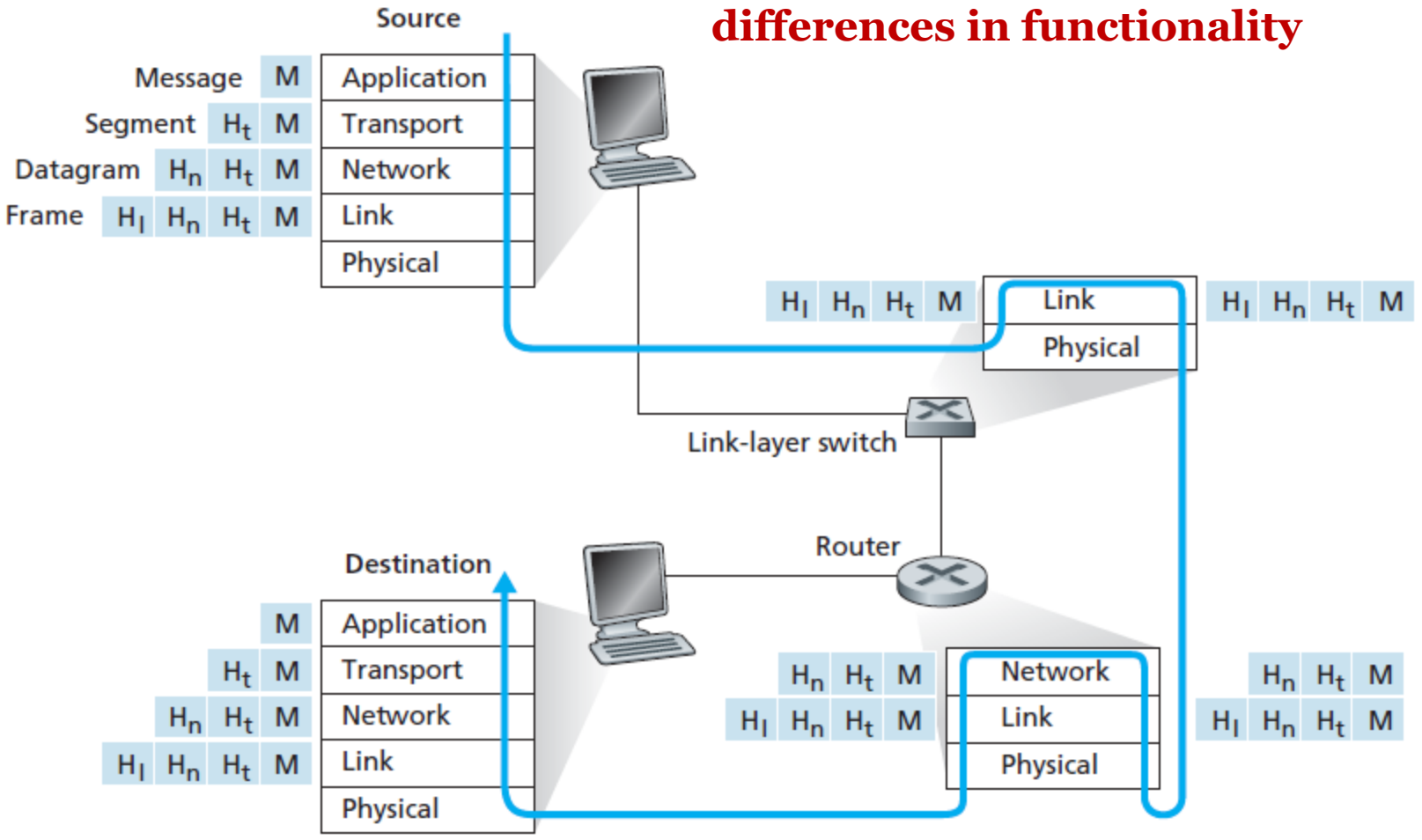
Internet protocol stack

- *application:* supporting network applications
  - FTP, SMTP, HTTP, IMAP, DNS
- *transport:* process-process data transfer
  - TCP, UDP
- *network:* routing of datagrams from source to destination
  - IP, routing protocols
- *link:* data transfer between neighboring network elements
  - Ethernet, 802.111 (WiFi), PPP
- *physical:* bits "on the wire"



TCP/IP Model

Hosts, routers, and link-layer switches; each contains a different set of layers, reflecting their differences in functionality



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1.7 history

Network security

- Field of network security:
  - how bad guys can attack computer networks
  - how we can defend networks against attacks
  - how to design architectures that are immune to attacks
- Internet not originally designed with (much) security in mind
  - *original vision:* "a group of mutually trusting users attached to a transparent network" ☺
  - Internet protocol designers playing "catch-up"
  - security considerations in all layers!

Bad guys: put malware into hosts via Internet

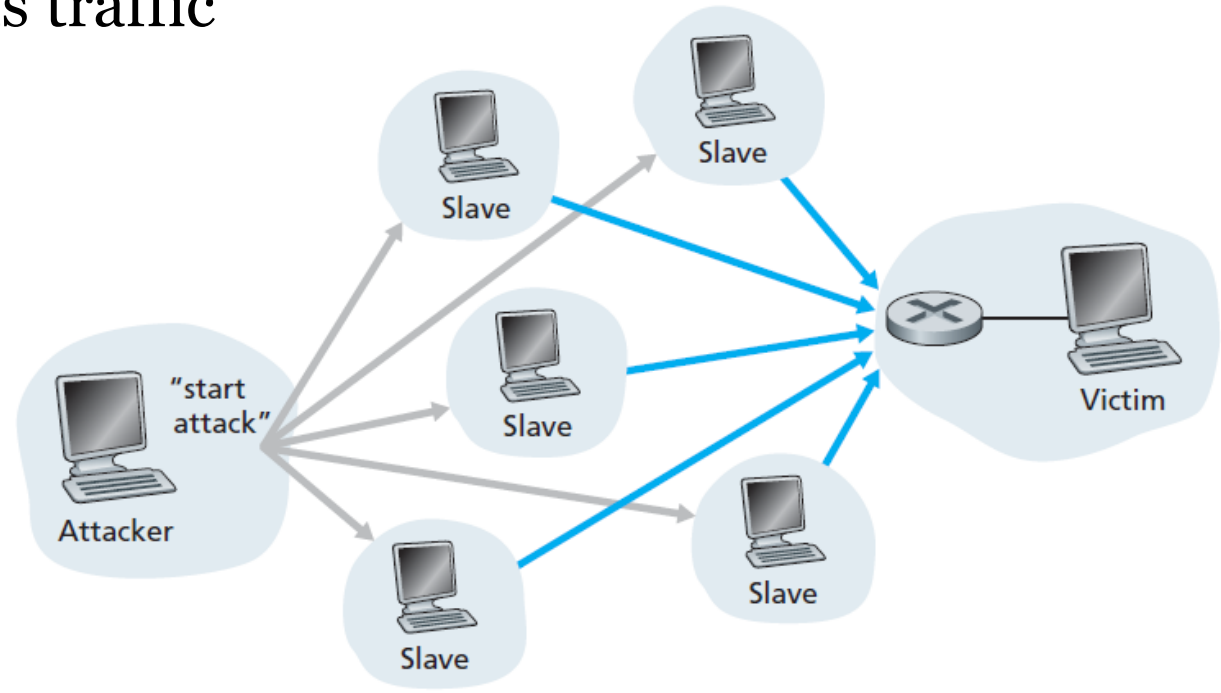
- Malware = malicious software
  - Examples: Worm, Virus, Trojans, spyware, botnet, etc.
- Malware can be
  - Self-replicating ➔ once it infects one host, from that host it seeks entry into other hosts over the Internet, and just keep going
  - User activated (click on an email attachment) or
  - Program activated (a vulnerable app that activates malware)
  - Recruiting your device to be part of a botnet, to be use for any purpose later

Bad guys: put malware into hosts via Internet

- malware can get in host from:
  - *virus:* self-replicating infection by receiving/executing object (e.g., e-mail attachment)
  - *worm:* self-replicating infection by passively receiving object that gets itself executed
- spyware malware can record keystrokes, web sites visited, upload info to collection site
- infected host can be enrolled in botnet, used for spam. DDoS attacks

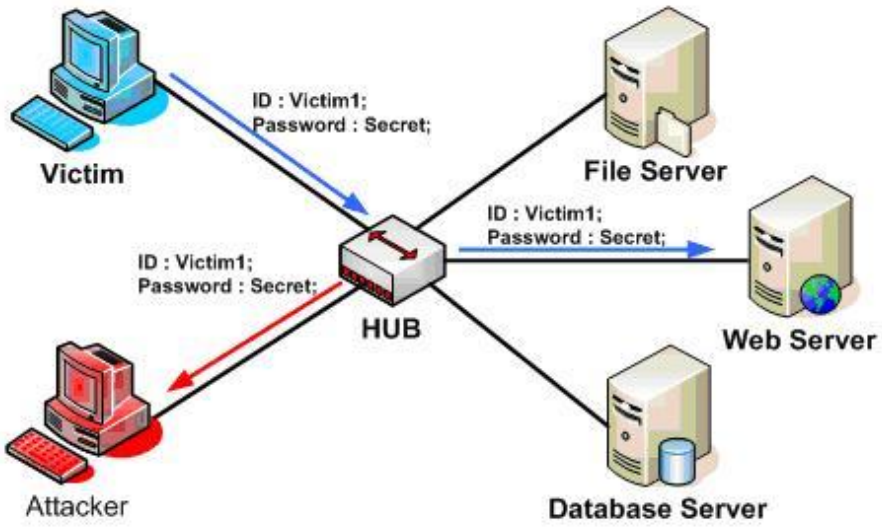
Denial of Service (DoS): attackers make resources (server, bandwidth) unavailable to legitimate traffic by overwhelming resource with bogus traffic

1. select target
2. break into hosts around the network (see botnet)
3. send packets to target from compromised hosts



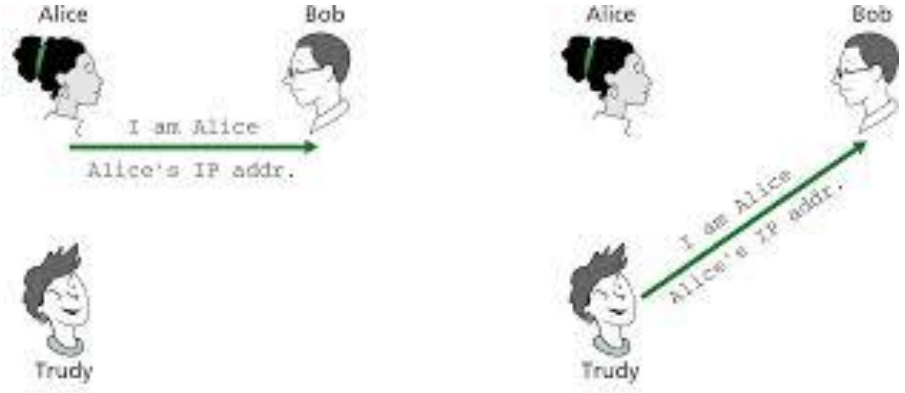
Packet "sniffing":

- broadcast media (shared Ethernet, wireless)
- promiscuous network interface reads/records all packets (e.g., including passwords!) passing by

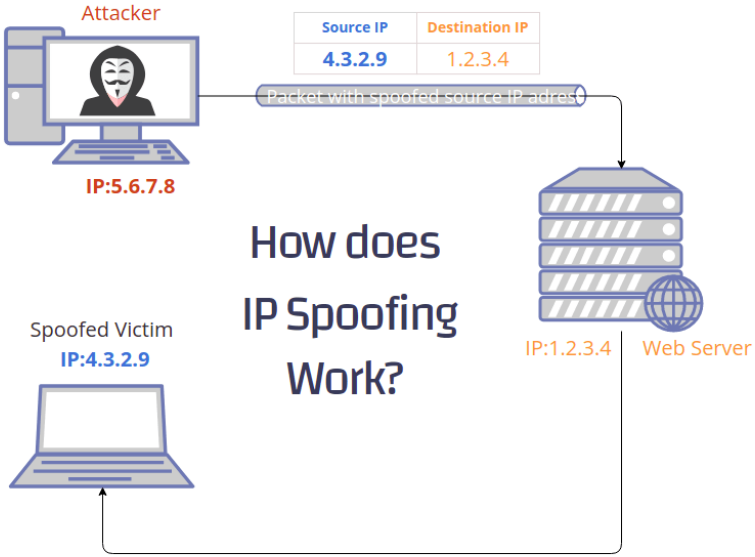


Wireshark software used for end-of-chapter labs is a (free) packet-sniffer

IP spoofing: send packet with false source address → injection of packet with false source address



- Protocol *ap2.0* and a failure scenario



Lines of defense:

- authentication: proving you are who you say you are
  - cellular networks provides hardware identity via SIM card; no such hardware assist in traditional Internet
- confidentiality: via encryption
- integrity checks: digital signatures prevent/detect tampering
- access restrictions: password-protected VPNs
- firewalls: specialized "middleboxes" in access and core networks:
  - off-by-default: filter incoming packets to restrict senders, receivers, applications
  - detecting/reacting to DOS attacks

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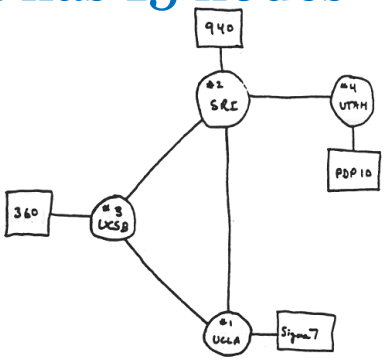
1.7 history

Internet history

1961-1972: Early packet-switching principles

- 1961: Kleinrock - queueing theory shows effectiveness of packet-switching
- 1964: Baran - packet-switching in military nets
- 1967: ARPAnet conceived by Advanced Research Projects Agency
- 1969: first ARPAnet node operational

- 1972:
  - ARPAnet public demo
  - NCP (Network Control Protocol) first host-host protocol
  - first e-mail program
  - ARPAnet has 15 nodes



THE ARPA NETWORK

Internet history

1972-1980: Internetworking, new and proprietary nets

- 1970: ALOHAnet satellite network in Hawaii
- 1974: Cerf and Kahn - architecture for interconnecting networks
- 1976: Ethernet at Xerox PARC
- late70's: proprietary architectures: DECnet, SNA, XNA
- late 70's: switching fixed length packets (ATM precursor)
- 1979: ARPAnet has 200 nodes

Cerf and Kahn's internetworking principles:

- minimalism, autonomy - no internal changes required to interconnect networks
- best effort service model
- stateless routers
- decentralized control

define today's Internet architecture

Internet history

1980-1990: new protocols, a proliferation of networks

- 1983: deployment of TCP/IP
- 1982: smtp e-mail protocol defined
- 1983: DNS defined for name-to-IP-address translation
- 1985: ftp protocol defined
- 1988: TCP congestion control

- new national networks: Csnet, BITnet, NSFnet, Minitel
- 100,000 hosts connected to confederation of networks

Internet history

1990, 2000's: commercialization, the Web, new apps

- early 1990's: ARPAnet decommissioned
- 1991: NSF lifts restrictions on commercial use of NSFnet (decommissioned, 1995)
- early 1990s: Web
  - hypertext [Bush 1945, Nelson 1960's]
  - HTML, HTTP: Berners-Lee
  - 1994: Mosaic, later Netscape
  - late 1990's: commercialization of the Web

late 1990's – 2000's:

- more killer apps: instant messaging, P2P file sharing
- network security to forefront
- est. 50 million host, 100 million+ users
- backbone links running at Gbps

2005-present : scale, SDN, mobility, cloud

- Aggressive deployment of broadband home access(10-100's Mbps)
- 2008: *software-defined networking (SDN)*
- Increasing ubiquity of high-speed wireless access: 4G/5G, WiFi
- Service providers (Google, FB, Microsoft) create their own networks
  - Bypass commercial Internet, providing "instantaneous" access to search, email, social media , content etc.
- E-commerce, universities, enterprises running their services in "cloud" (eg, Amazon Web Services, Microsoft Azure)
- rise of smartphones: more mobile than fixed devices on Internet (2017)
- ~18B devices attached to Internet (2017)

Introduction: summary

Covered a "ton" of material!

- Internet overview
- what's a protocol?
- network edge, core, access network
  - packet-switching versus circuit-switching
  - Internet structure
- performance: loss, delay, throughput
- layering, service models
- security
- history

You now have:

- context, overview, "feel" of networking
- more depth, detail *to follow!*

Homework 1

Individual:

- Do a mind map of CH1.
  - Start on first class, then add to it
  - Submit via e-learning.

- Go through the history videos.
  - May be asked in exam